CME 250:
Introduction to Machine Learning

Lecture 7:
Unsupervised Learning

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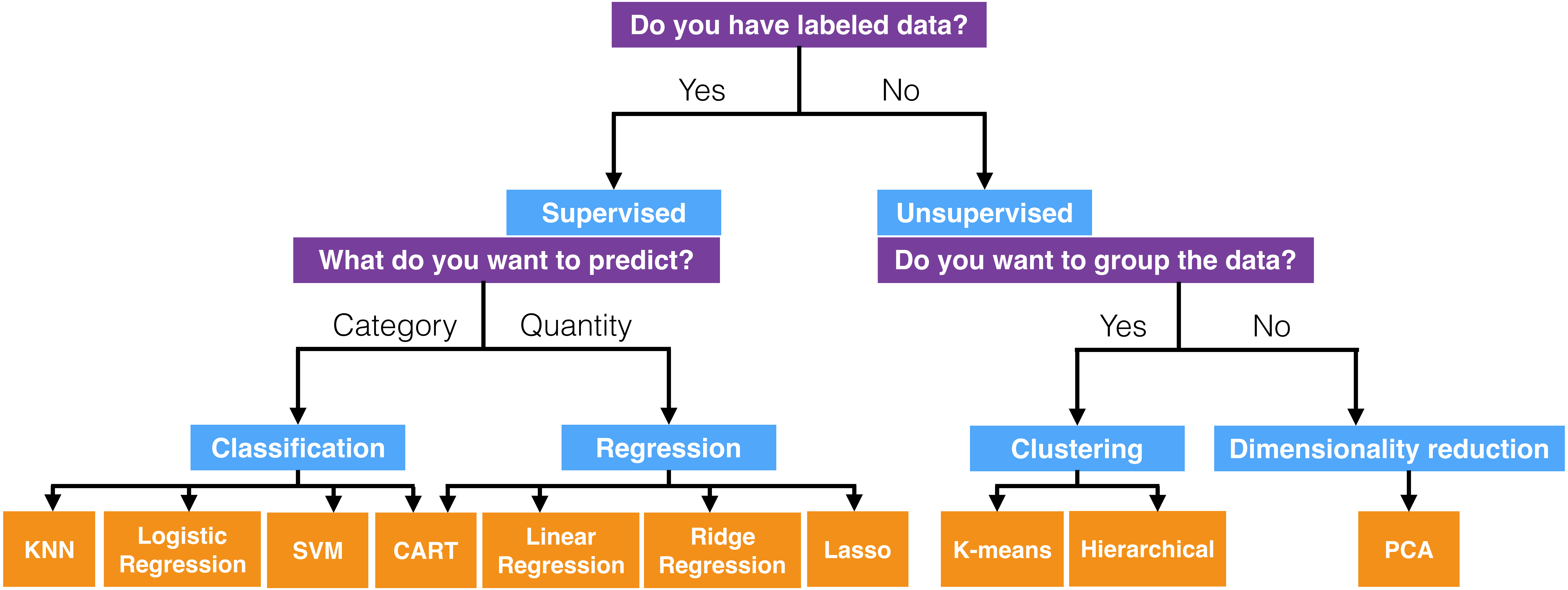
Agenda

Slides are online at cme250.stanford.edu

- Clustering methods
  - K-means clustering
  - Hierarchical clustering
- Dimensionality reduction
  - PCA

Machine Learning Methods



Unsupervised Learning

Recall: A set of statistical tools for data that only has features/input available, but no response.

In other words, we have X's but no labels y.

Goal: Discover interesting patterns/properties of the data.

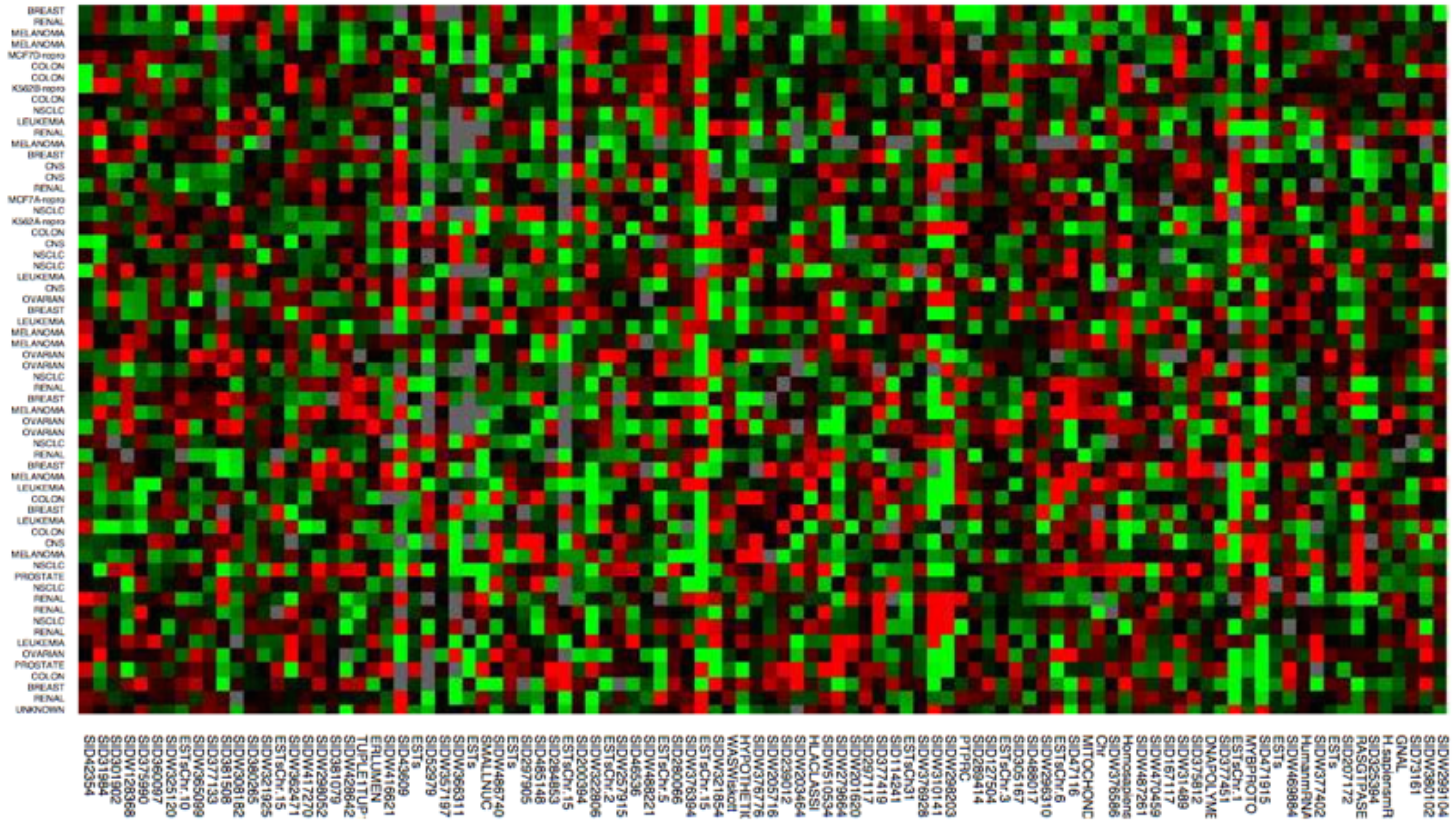
- E.g. for visualizing or interpreting high-dimensional data.

Unsupervised Learning

Example applications:

- Given tissue samples from n patients with breast cancer, identify unknown subtypes of breast cancer.

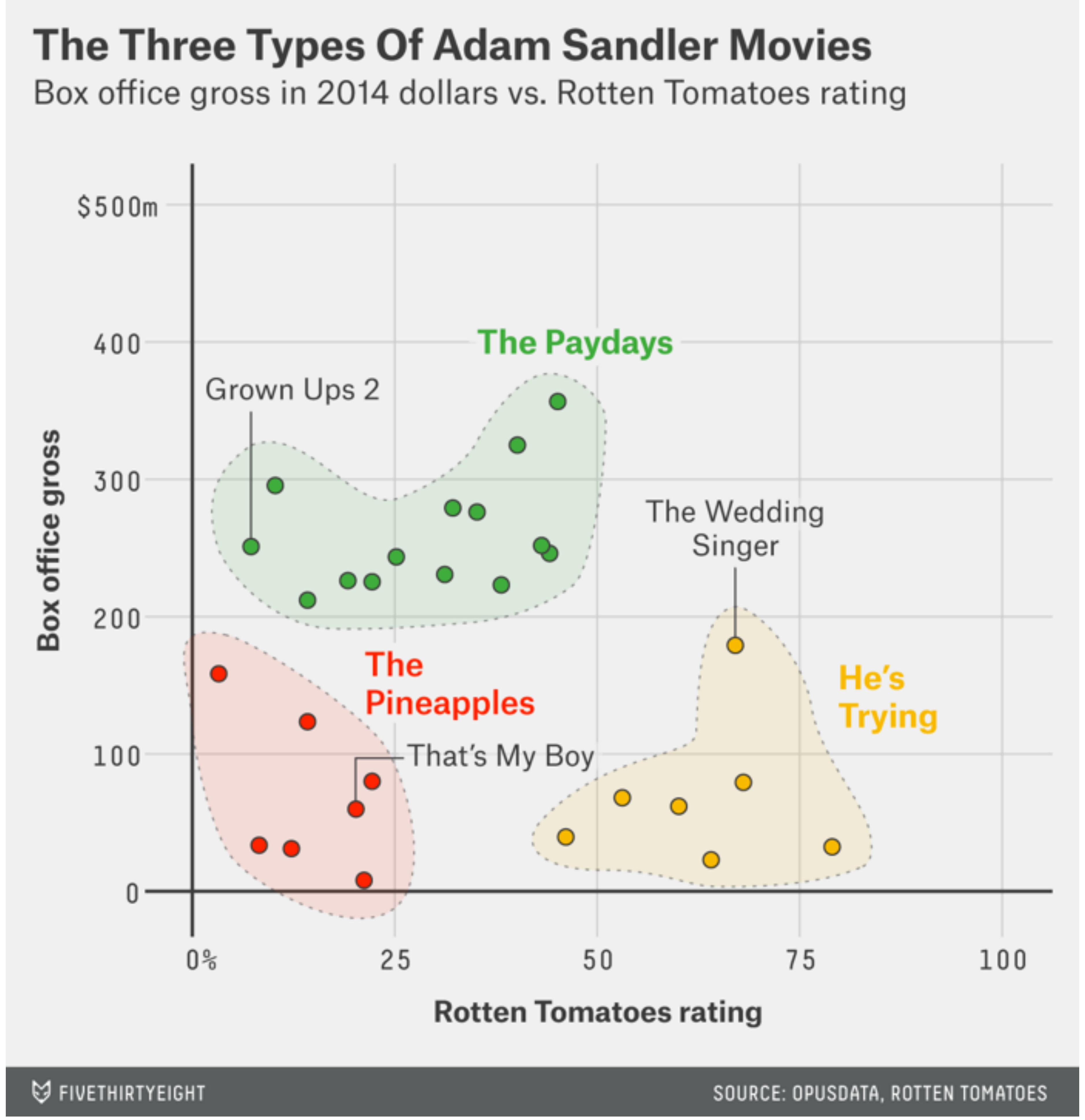
- Gene expression experiments have thousands of variables. Represent the data using a smaller set of features for visualization and interpretation.



Unsupervised Learning

Example applications:

- Document clustering: identify sets of documents about the same topic.
- Given high-dimensional facial images, find a compact representation as inputs for a facial recognition classifier.



Challenges of Unsupervised Learning

Why is unsupervised learning challenging?

- Exploratory data analysis — goal is not always clearly defined
- Difficult to assess performance — “right answer” unknown
- Working with high-dimensional data

Types of Unsupervised Learning

Two approaches:

- **Cluster analysis**
  - For identifying homogenous subgroups of samples
- **Dimensionality reduction**
  - For finding a low-dimensional representation to characterize and visualize the data

Cluster Analysis

Clustering

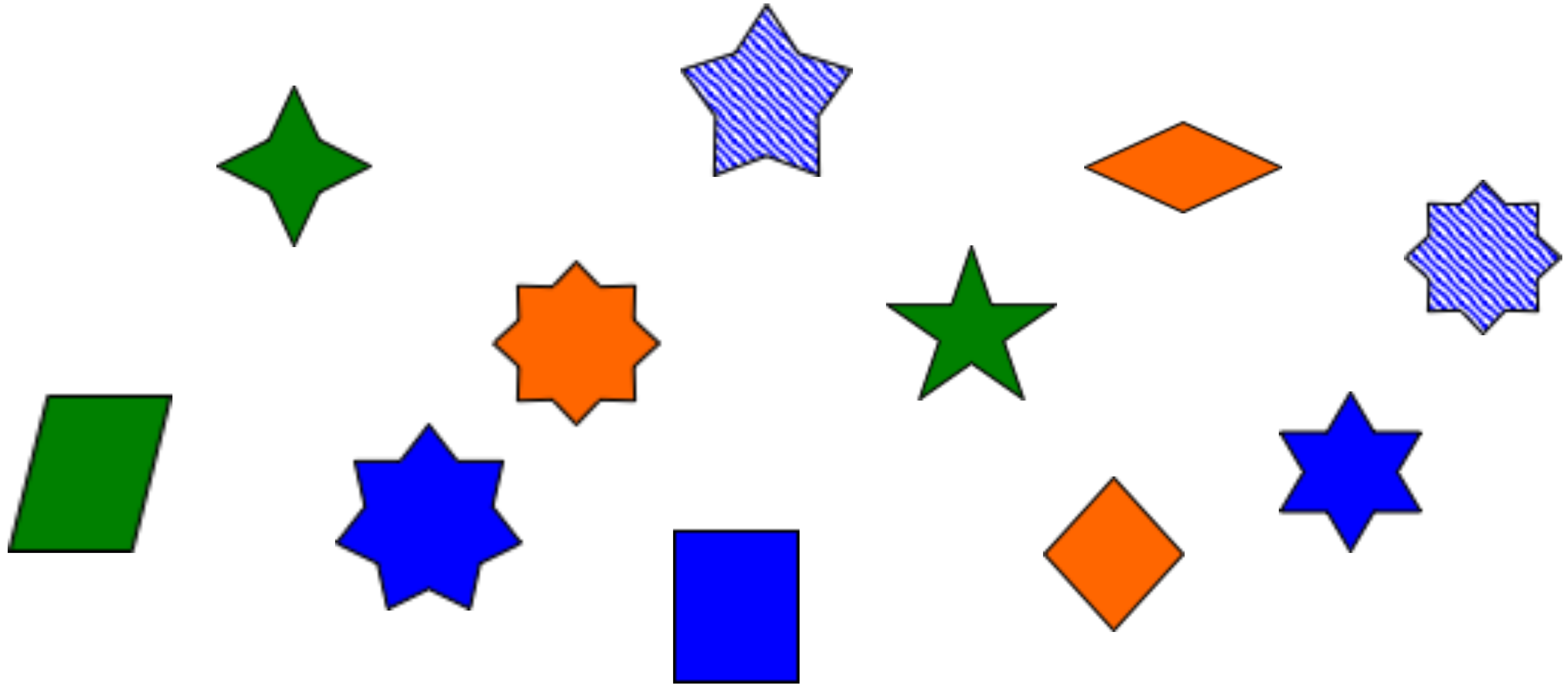
A set of methods for finding subgroups within the dataset.

- Observations should share common characteristics within the same group, but differ across groups.

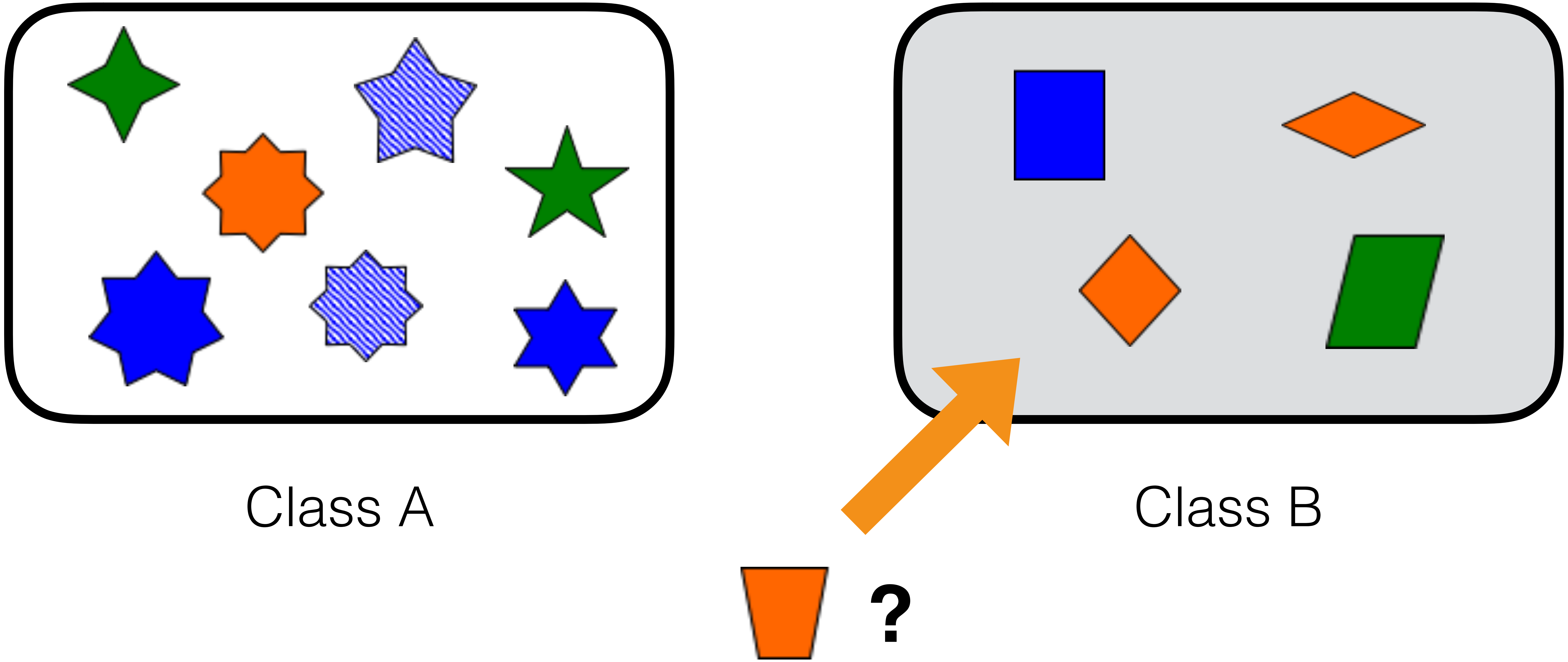
- Groupings are determined from attributes of the data itself — differs from classification.

https://medium.com/square-corner-blog/so-you-have-some-clusters-now-what-abfd297a575b

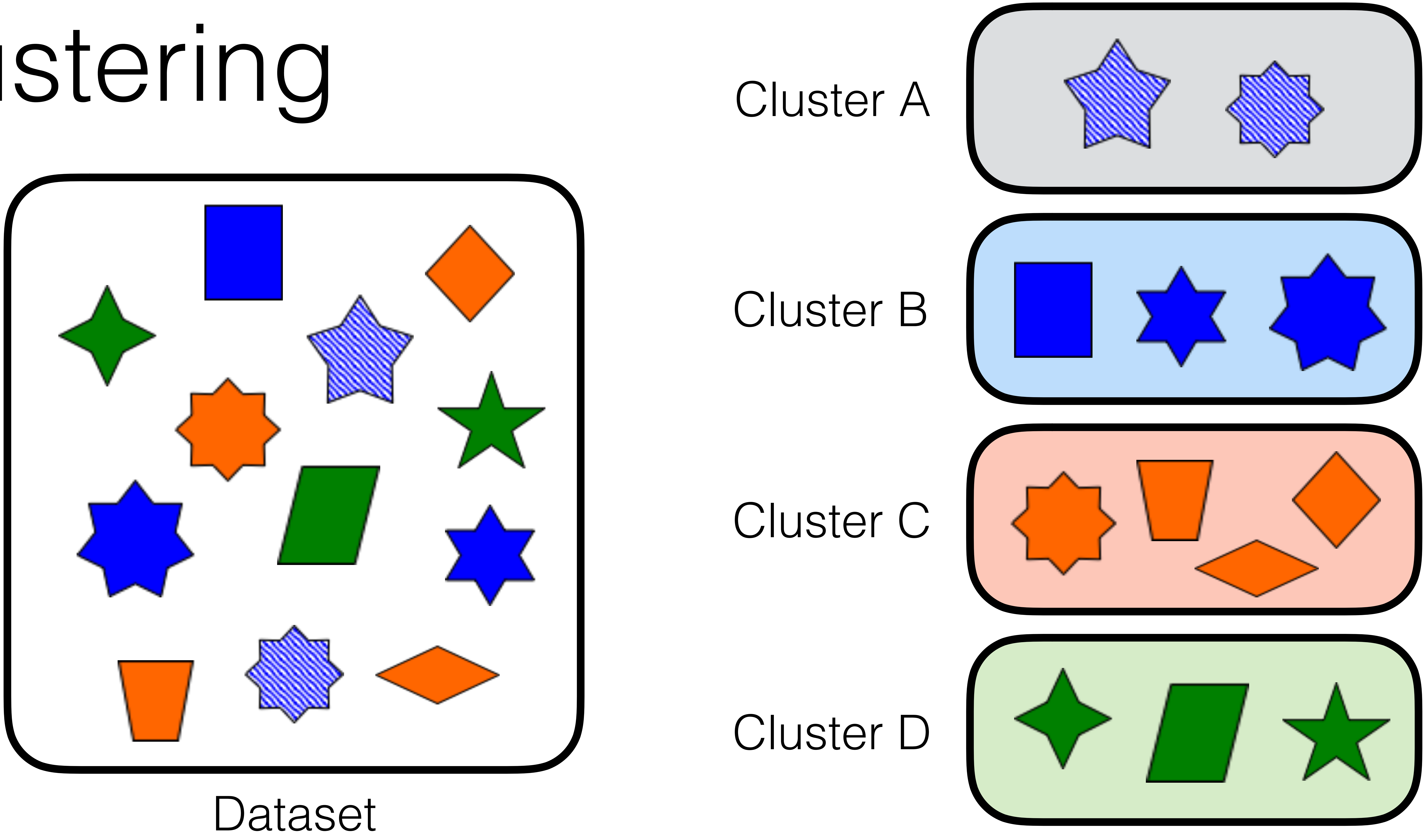
Clustering vs. Classification



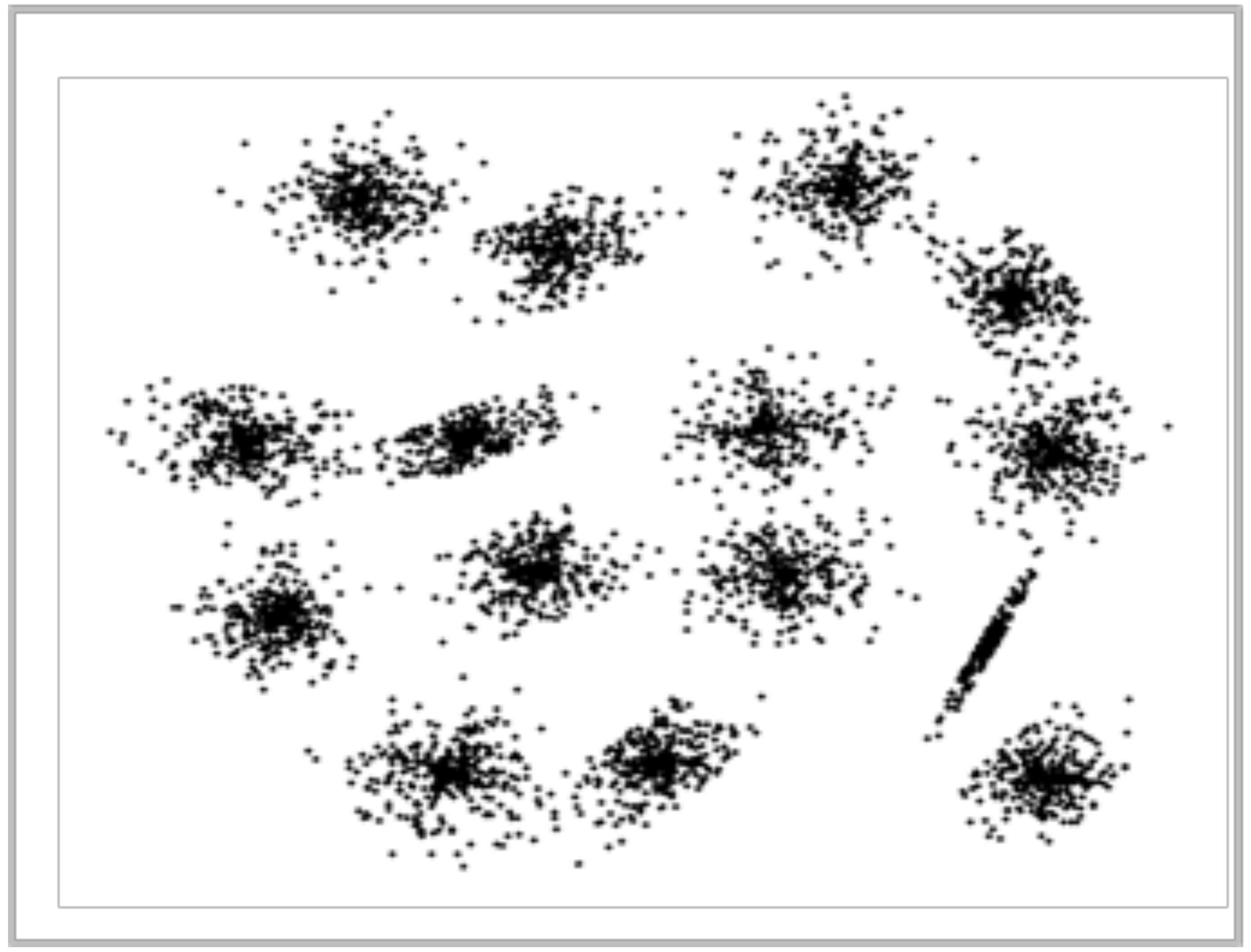
Classification



Clustering



Clustering



http://cs.joensuu.fi/sipu/datasets/

Types of Clustering

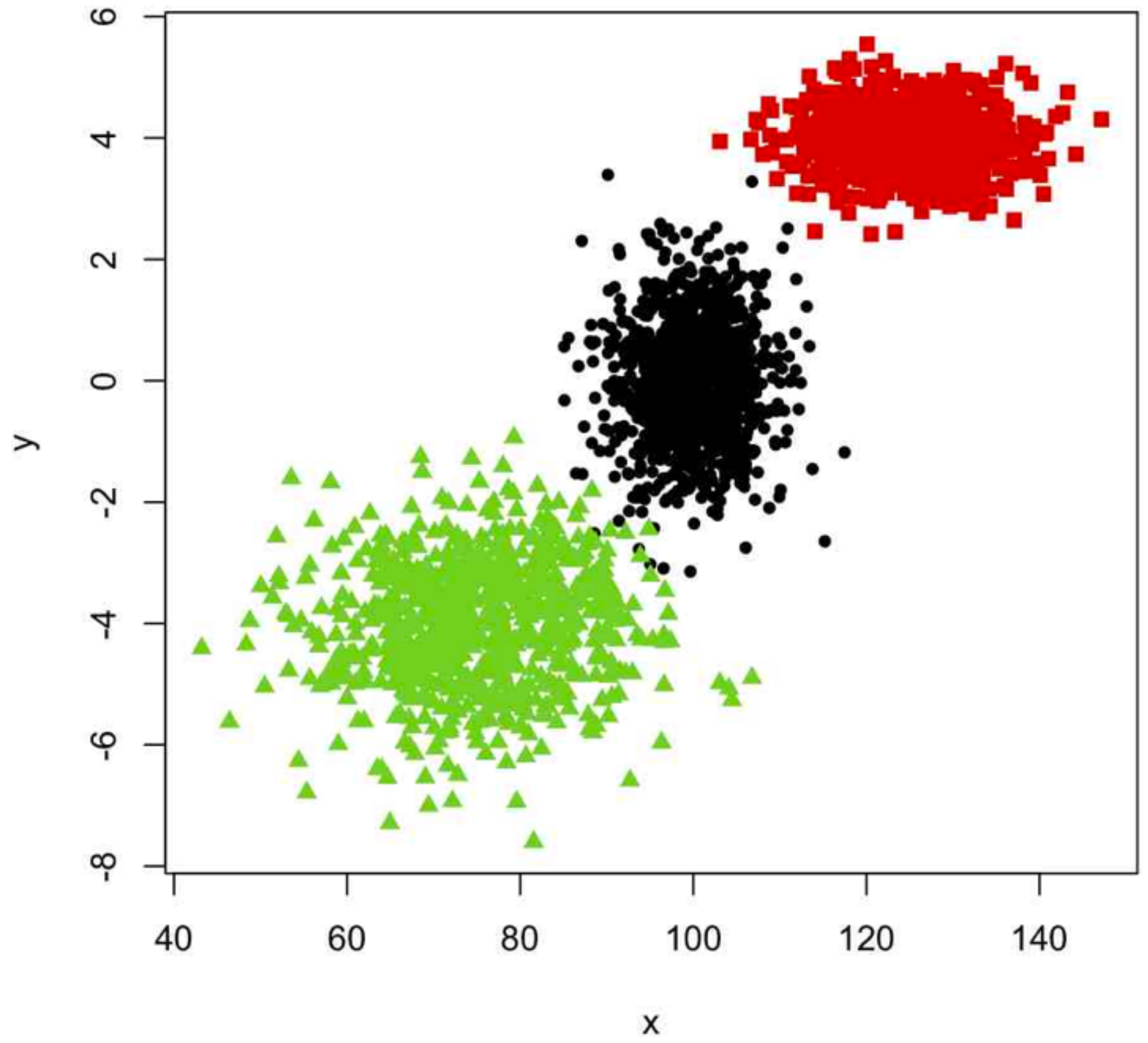
- **Centroid-based clustering**
- **Hierarchical clustering**
- **Model-based clustering**
  - Each cluster is represented by a parametric distribution
  - Dataset is a mixture of distributions
- **Hard vs. soft/fuzzy clustering**
  - Hard: observations divided into distinct clusters
  - Soft: observations may belong to more than one cluster

K-means Clustering

Groups data into K clusters that satisfy two properties.

1. Each observation belongs to at least one of the K clusters.

2. Clusters are non-overlapping. No observation belongs to more than one cluster.



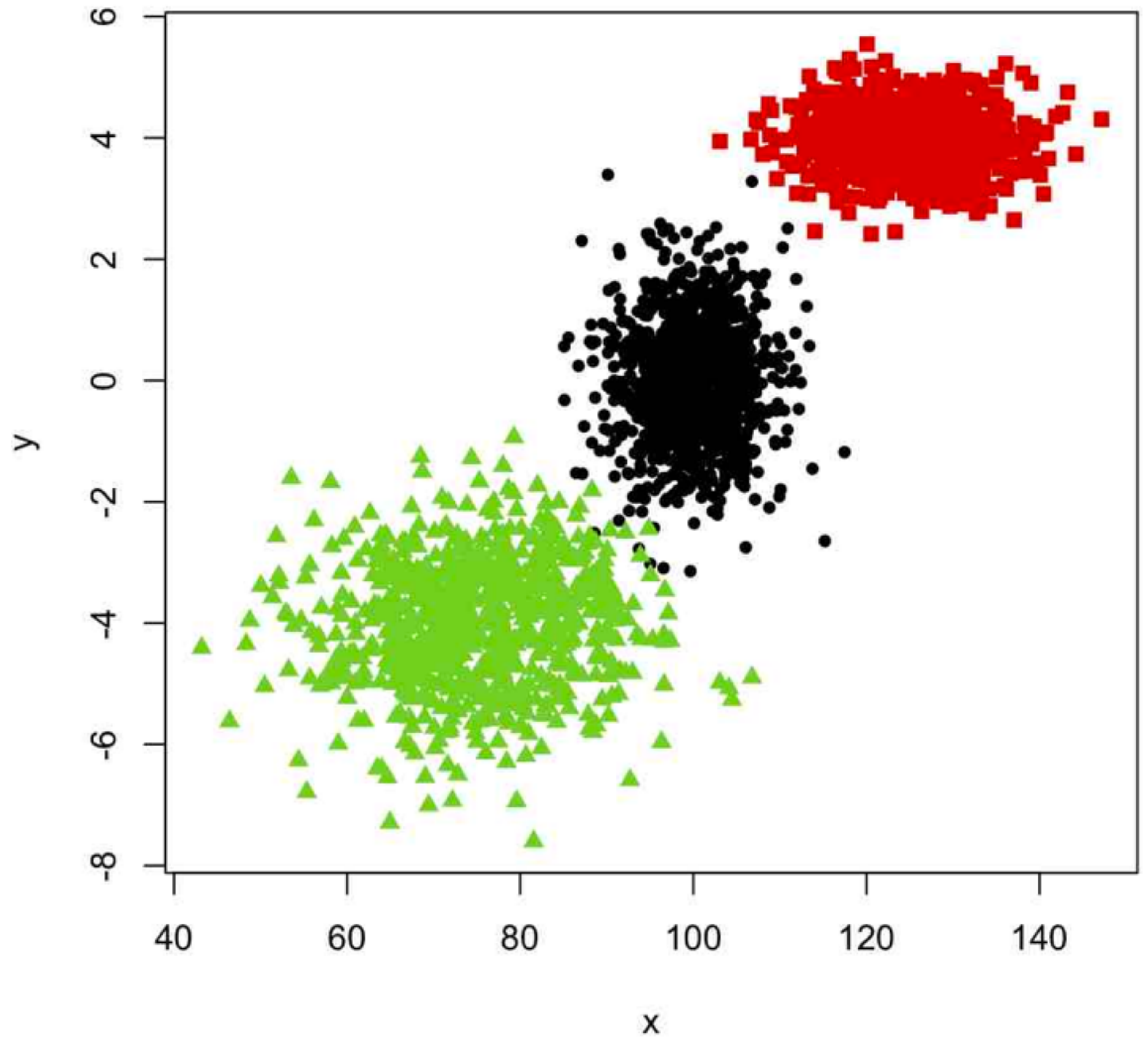
K-means Clustering

A good clustering is one for which the *within-cluster variation* is as small as possible.

Denote each cluster by C_k, and let $W(C_k)$ be a measure of the within-cluster variation.

K-means aims to solve

$$\underset{C_1,\ldots,C_K}{\text{minimize}} \left\{ \sum_{k=1}^{K} W(C_k) \right\}$$



K-means Clustering

How to measure within-cluster variation?

The most common choice is squared Euclidean distance.

$$W(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2$$

Which means overall we solve

$$\underset{C_1,\ldots,C_K}{\text{minimize}} \left\{ \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2 \right\}$$

K-means Clustering

It turns out that this optimization problem is difficult to solve, as it is discrete and there are nearly K^n ways to split n samples into K clusters.

In practice, use an iterative algorithm that finds a local minimum to this optimization.

K-means Clustering Algorithm

1. Initialize each observation to a cluster by randomly assigning a cluster, from 1 to K, to each observation.

2. Iterate until the cluster assignments stop changing:

   a. For each of the K clusters, compute the cluster centroid. The k-th cluster centroid is the vector of the p feature means for the observations in the k-th cluster.

   b. Assign each observation to the cluster whose centroid is closest (using Euclidean distance as the metric).

K-means Clustering Iterations

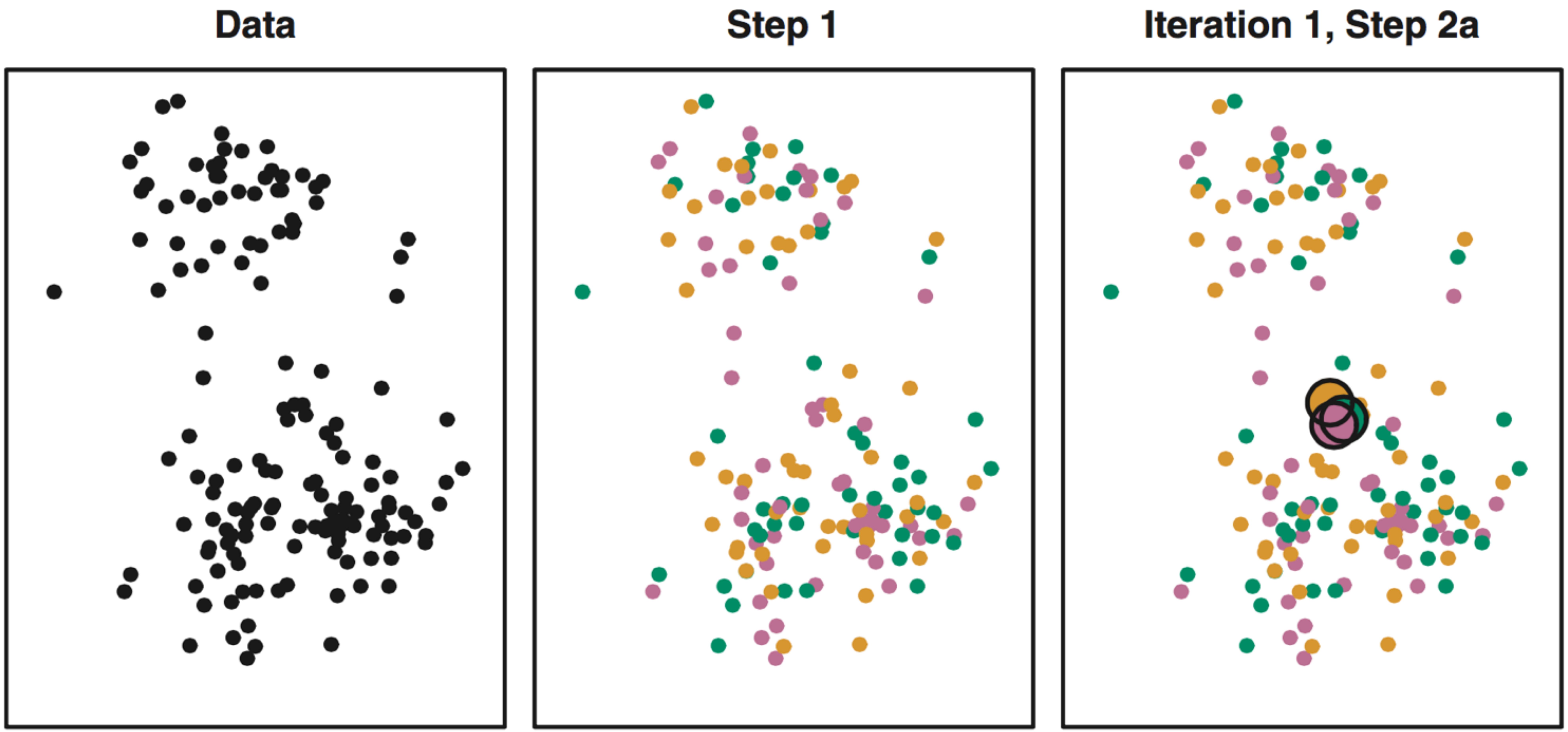


FIGURE 10.6, ISL (8th printing 2017)

K-means Clustering Iterations

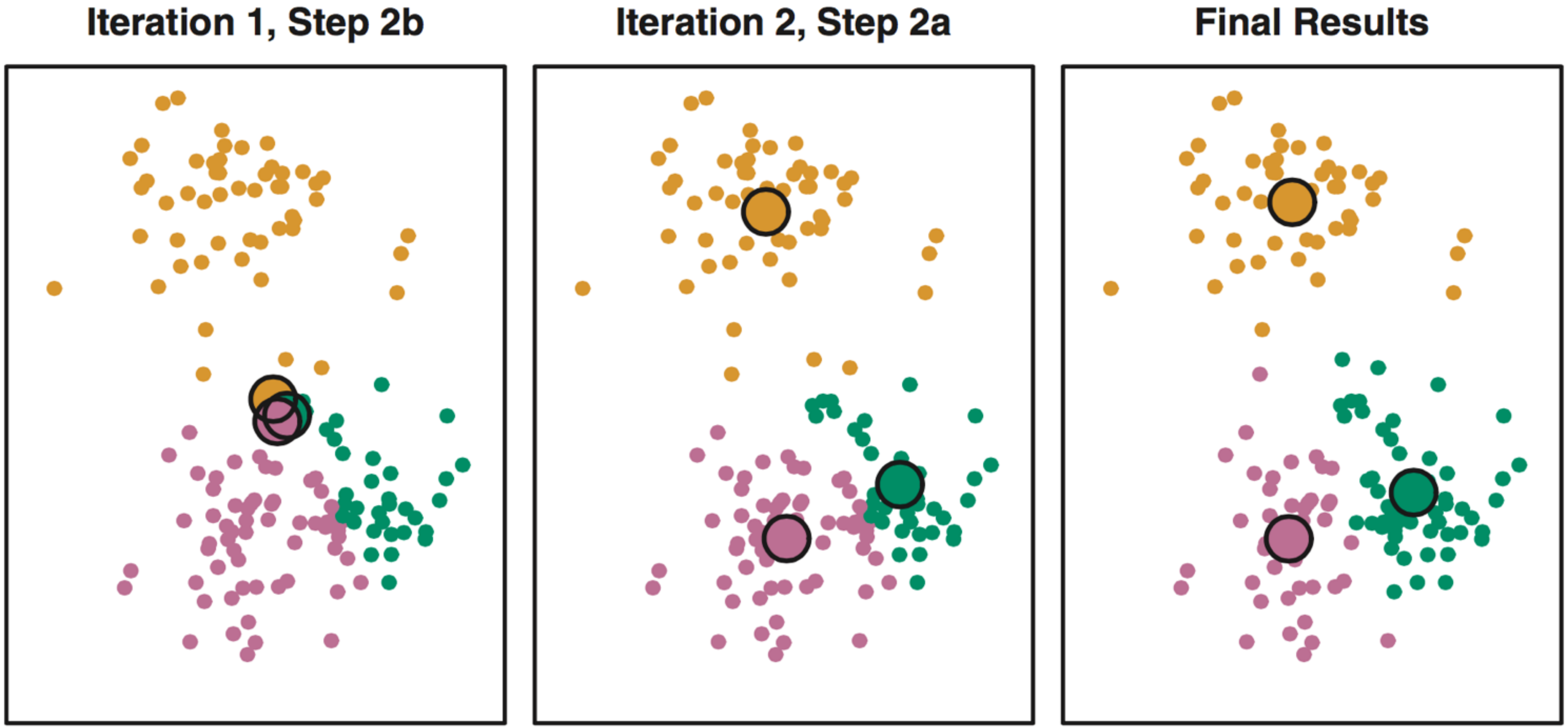
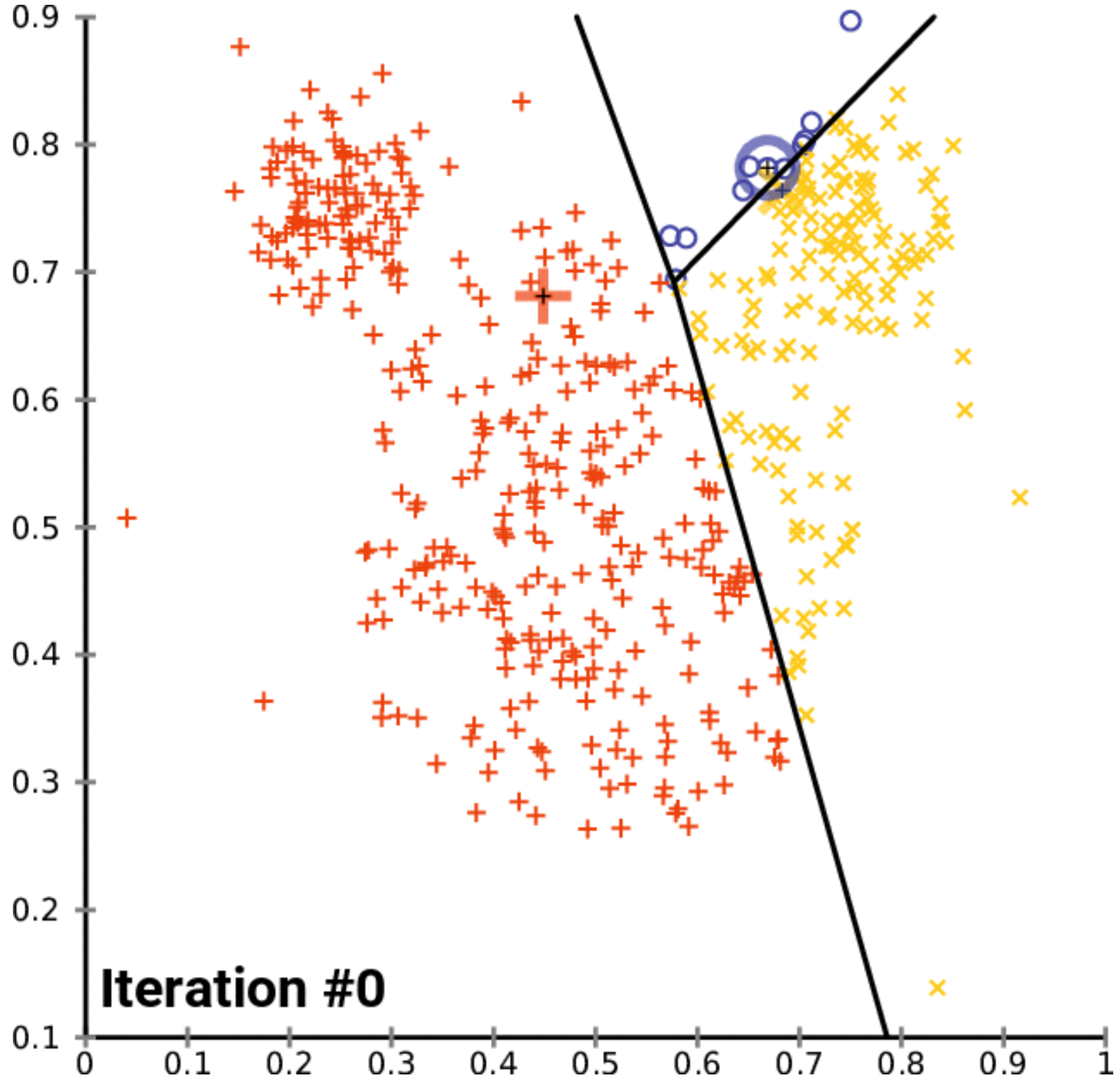


FIGURE 10.6, ISL (8th printing 2017)

K-means Clustering Animation



K-means Clustering Properties

It can be shown that the value of the objective function will never increase at each iteration of k-means.

Since the algorithm finds local minima, however, it will result in different clusters with different initializations.

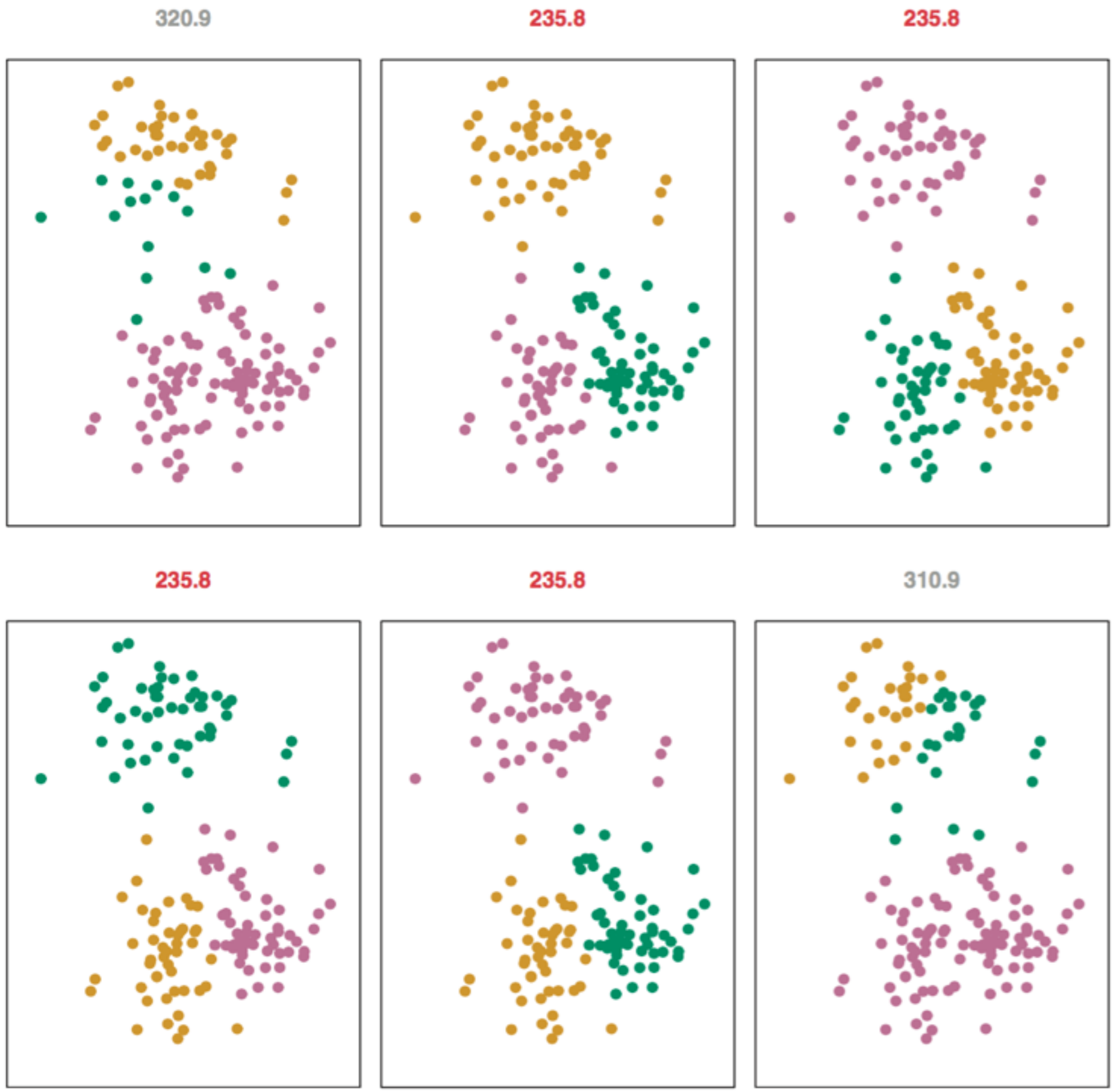


FIGURE 10.7, ISL (8th printing 2017)

K-means Pros and Cons

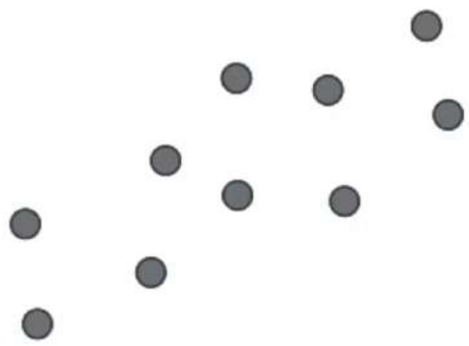
Pros:

- Easy to implement and understand

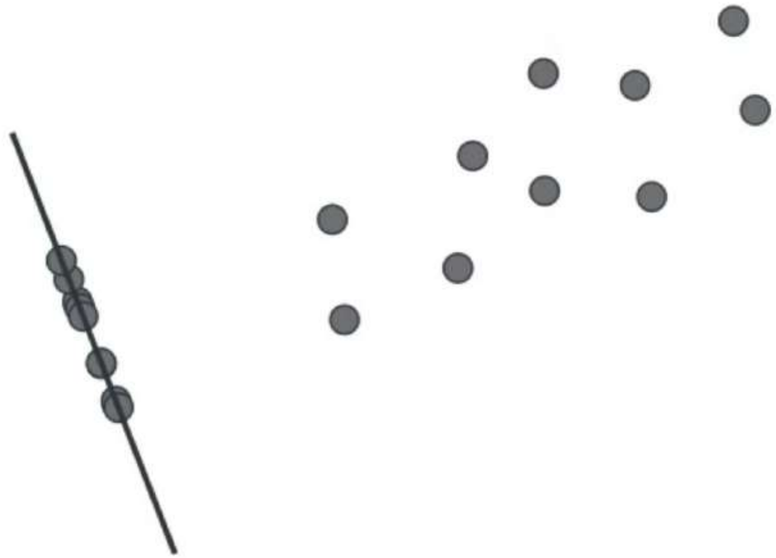
Cons:

- Not robust to data perturbations and different initializations
- Treats each feature equally, not robust to noise features or different scales of features — looks for in spherical clusters in feature space
- Need to define K before running algorithm

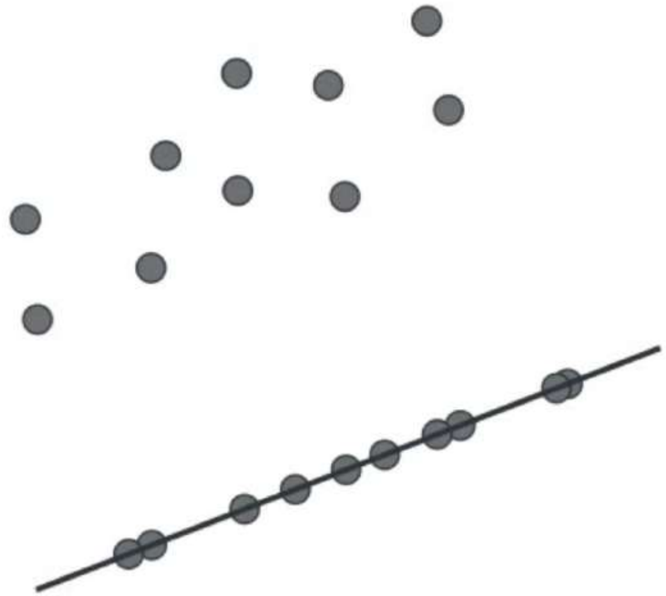
Dimensionality Reduction



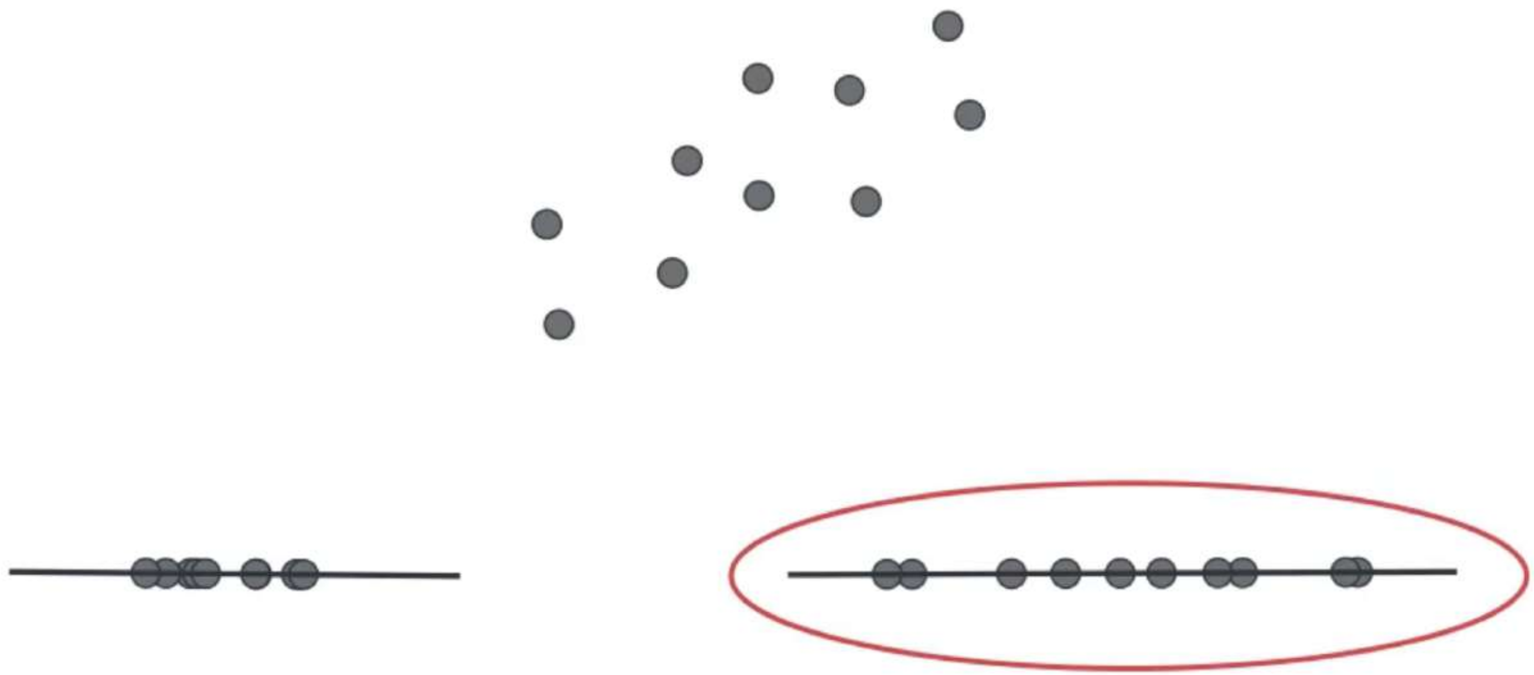
Dimensionality Reduction



Dimensionality Reduction



Dimensionality Reduction



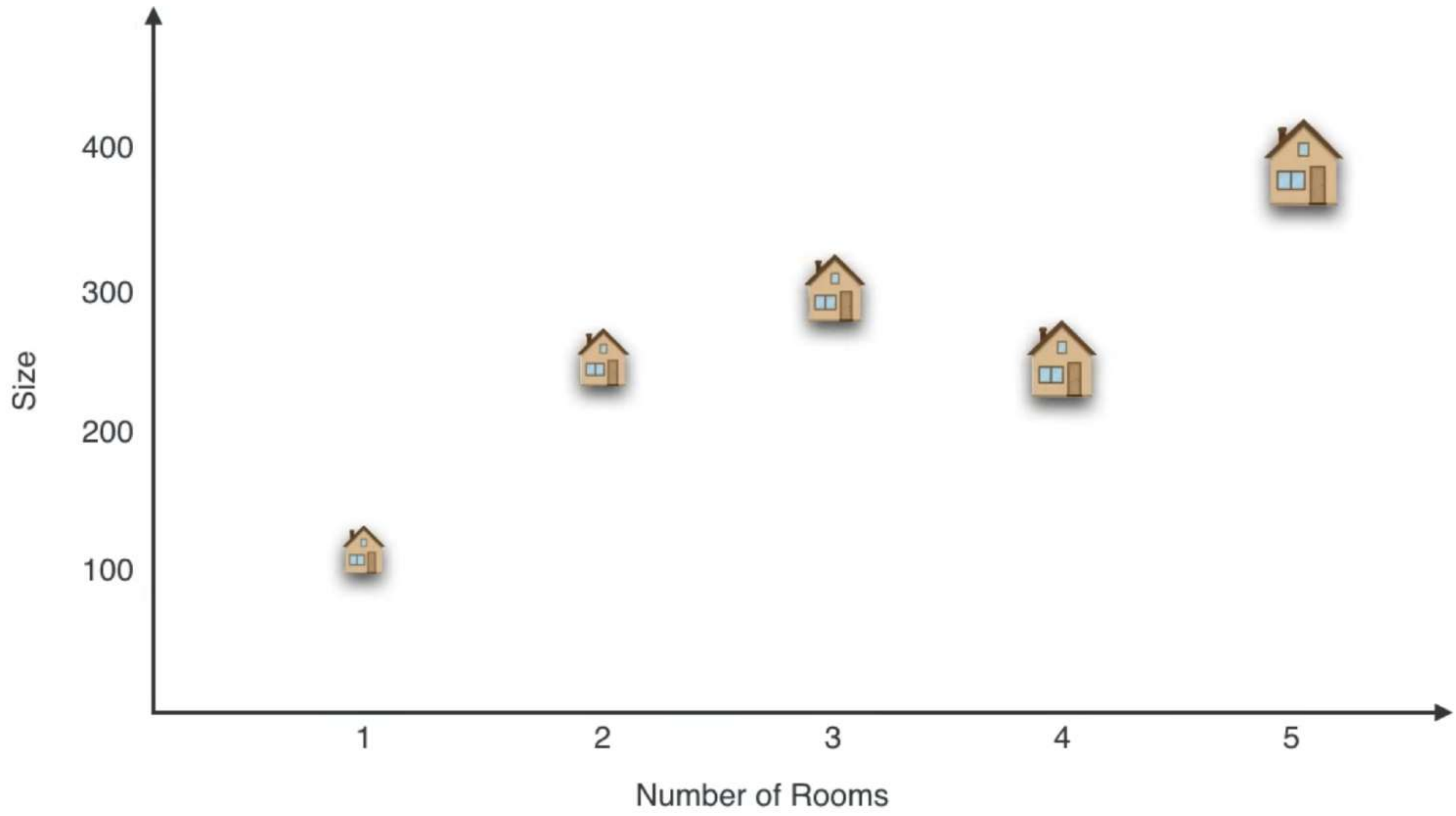
Housing Data

Size
Number of rooms
Number of bathrooms
Schools around
Crime rate

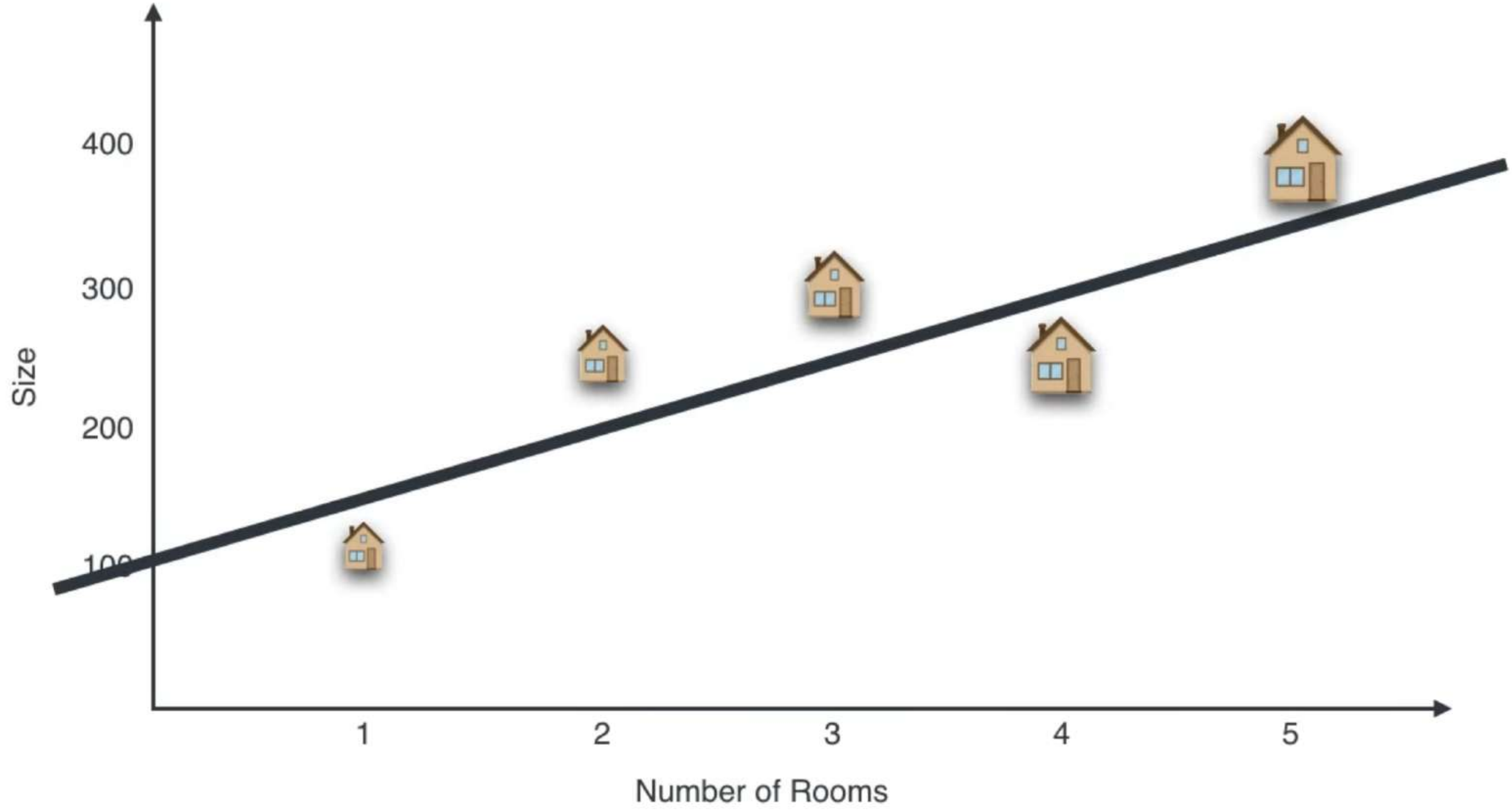
Housing Data

Size
Number of rooms → Size feature
Number of bathrooms

Schools around → Location feature
Crime rate



400
300
200
100
Size
1
2
3
4
5
Number of Rooms



400
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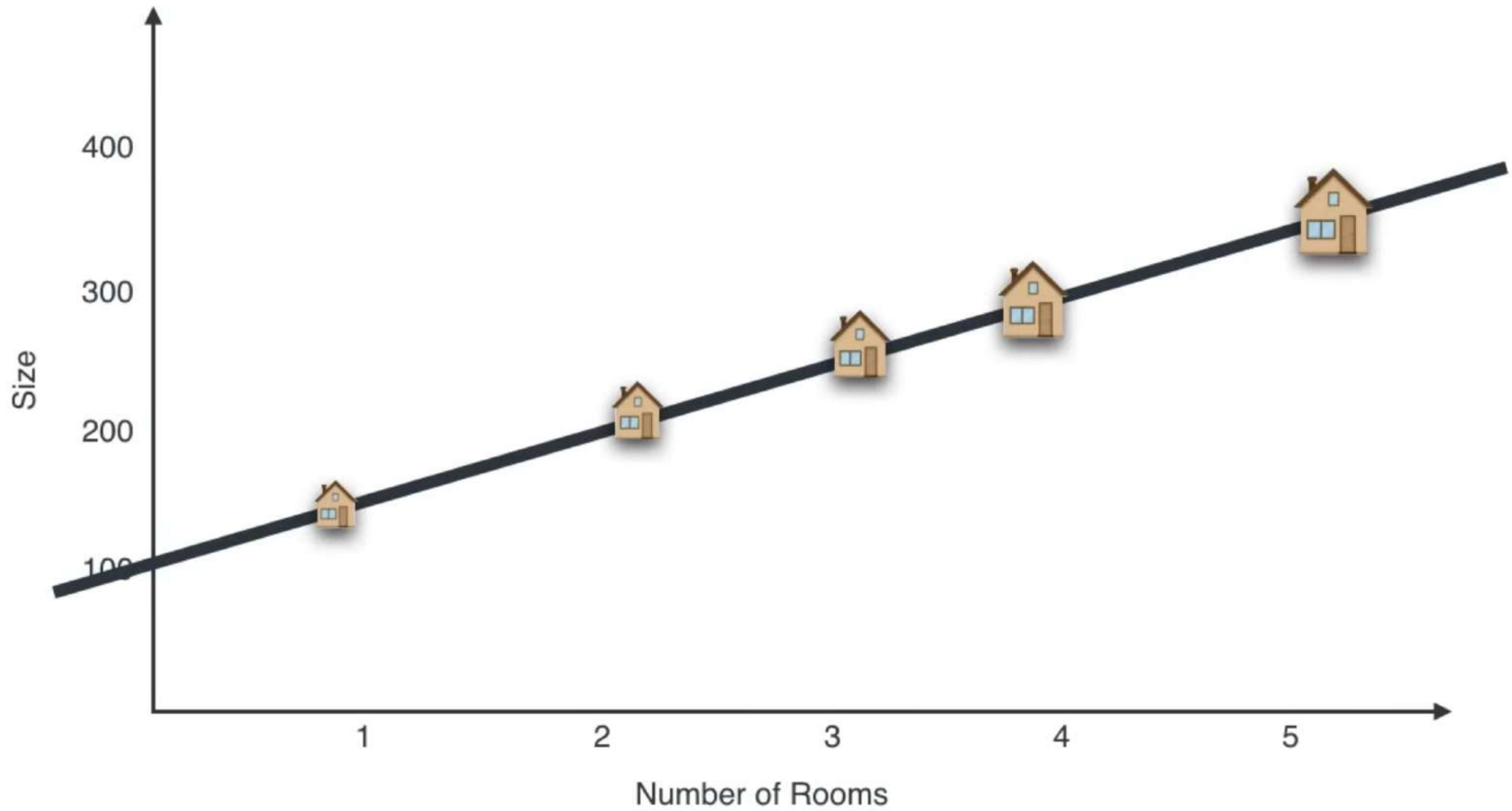

400
300
200
100
Size
1
2
3
4
5
Number of Rooms


Size feature

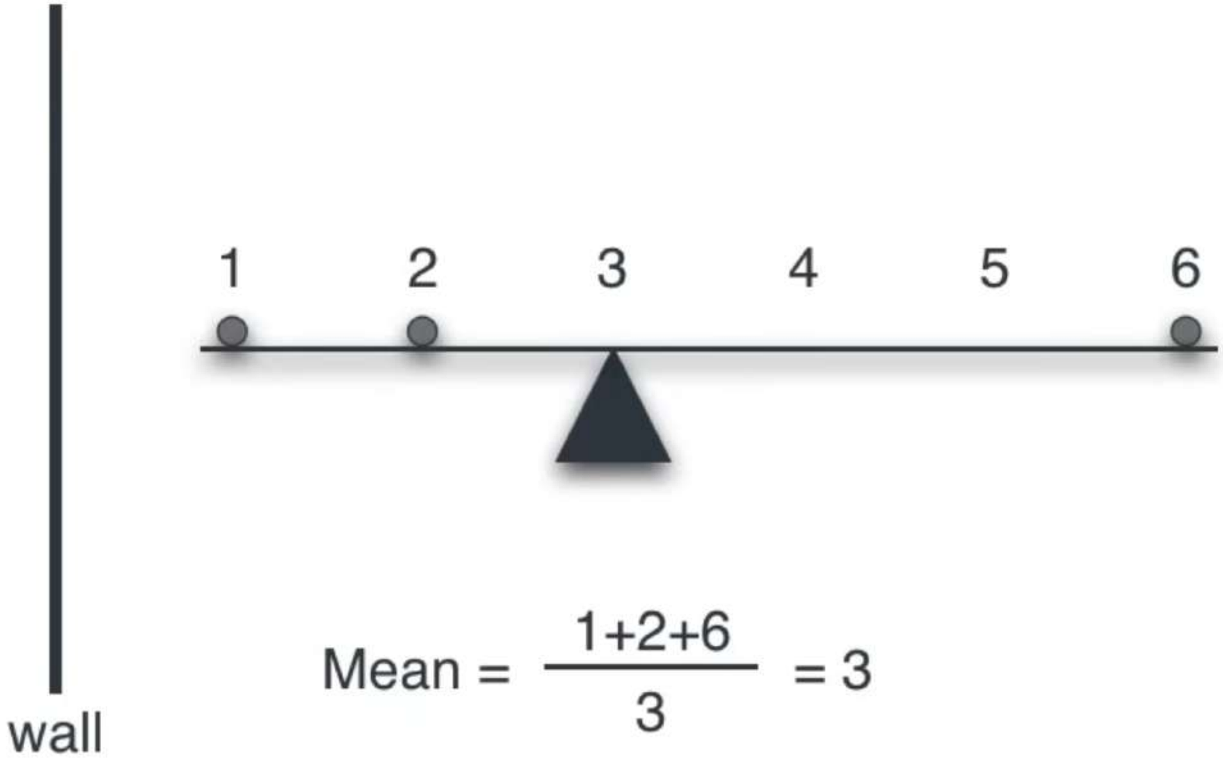
2 dimensions

size
number of rooms

1 dimension

size feature

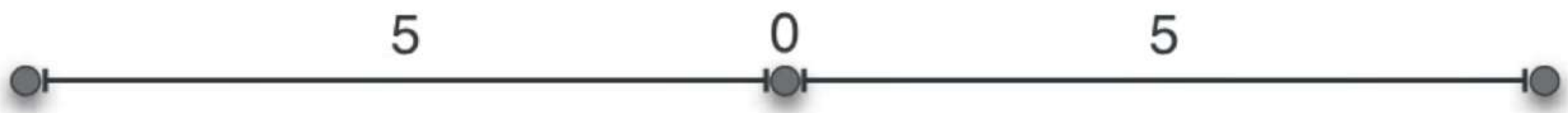
Mean



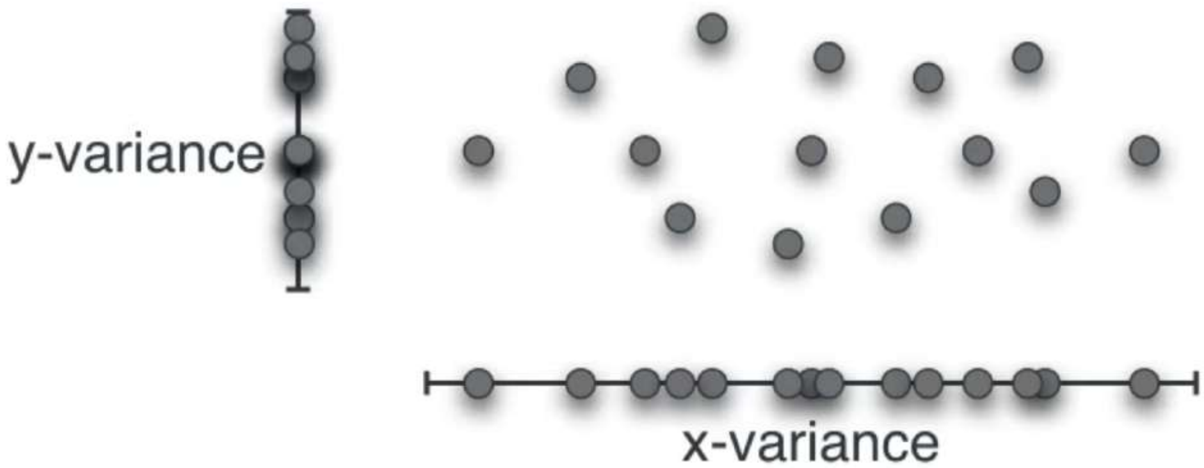
$$\text{Mean} = \frac{1+2+6}{3} = 3$$

Variance

$$\text{Variance} = \frac{1^2+0^2+1^2}{3} = 2/3$$



Variance?

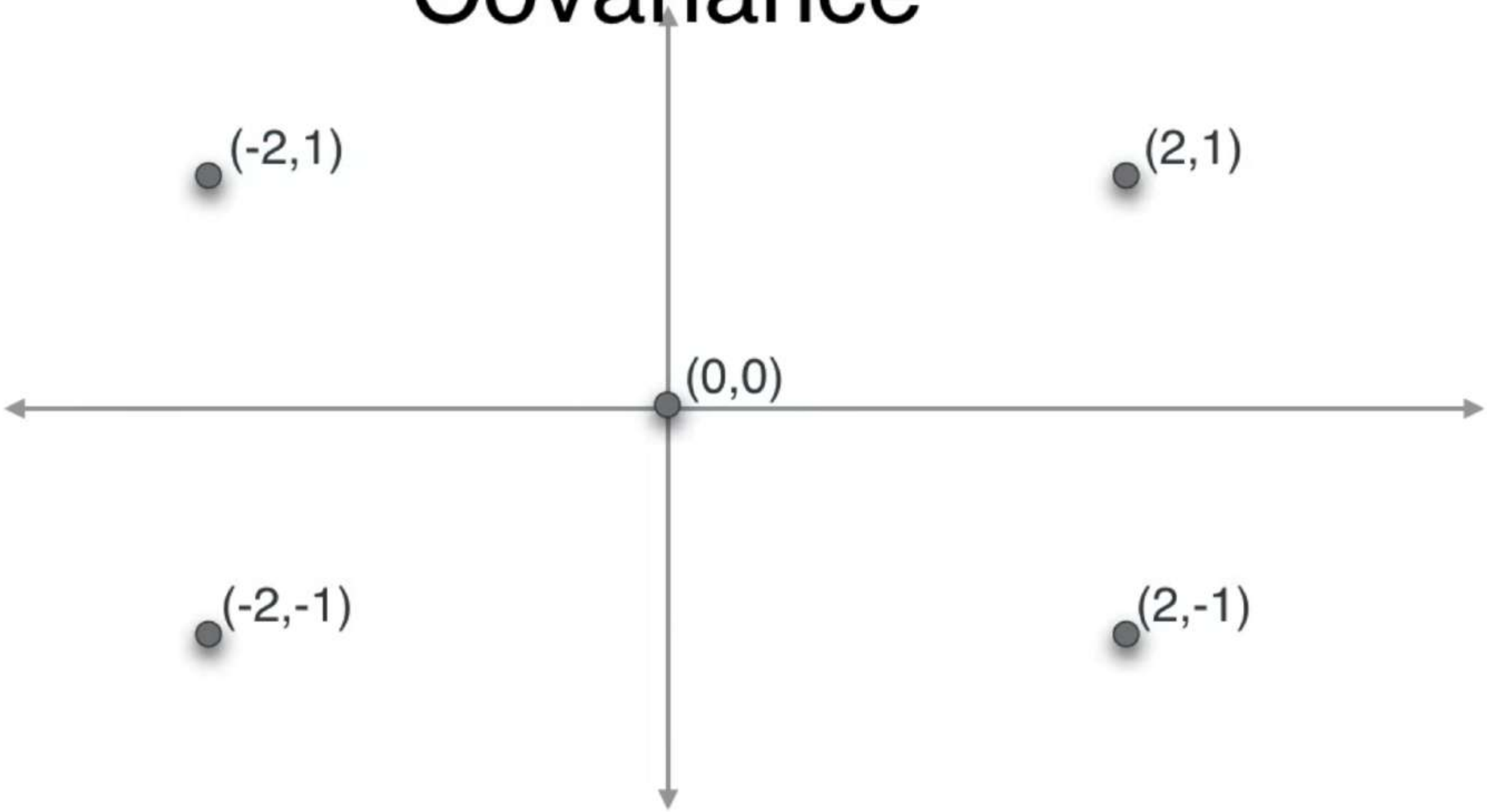


Variance?

$$\text{x-variance} = \frac{2^2+0^2+2^2}{3} = 8/3$$

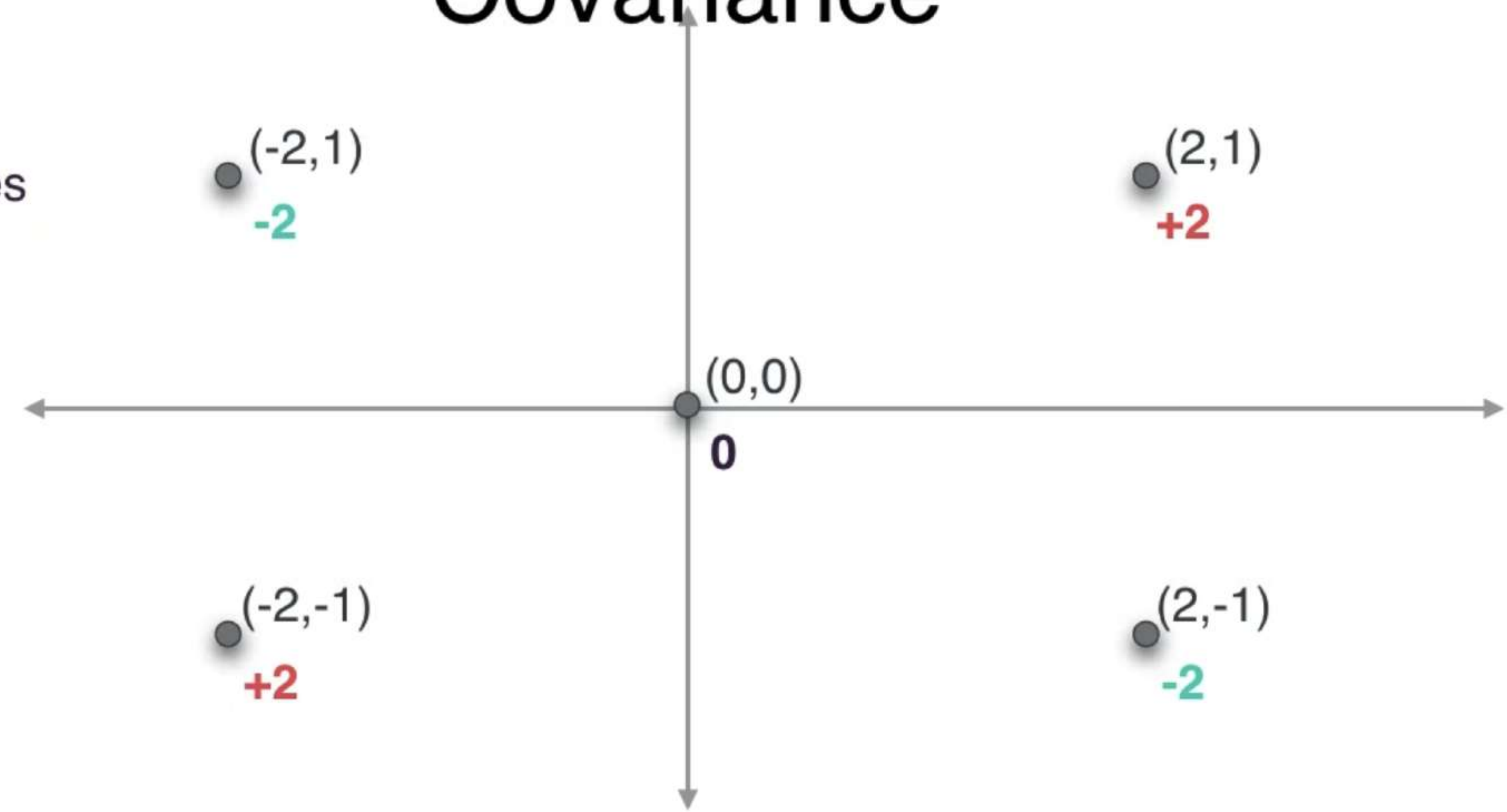
$$\text{y-variance} = \frac{1^2+0^2+1^2}{3} = 2/3$$

Covariance

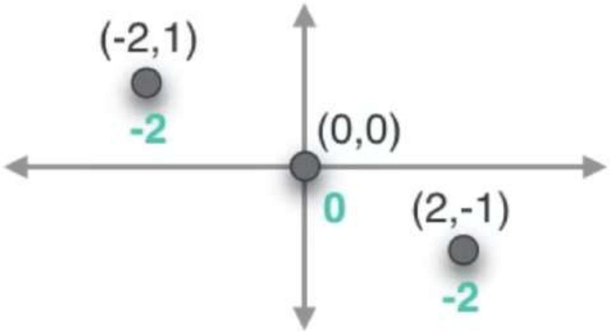


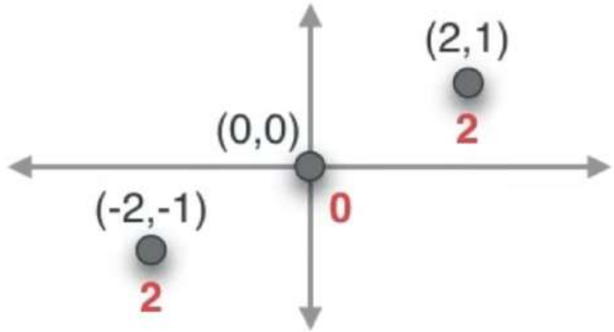
Covariance

Product of coordinates



Covariance

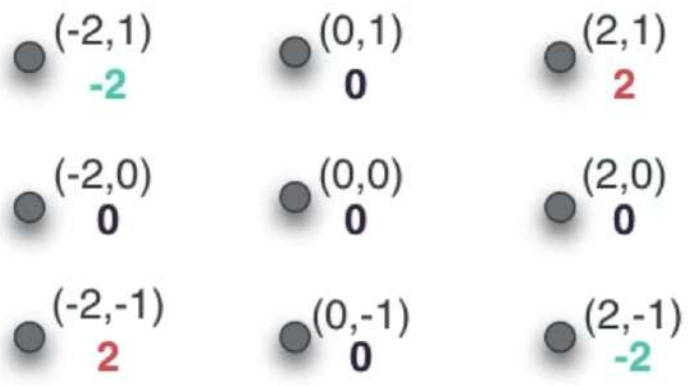




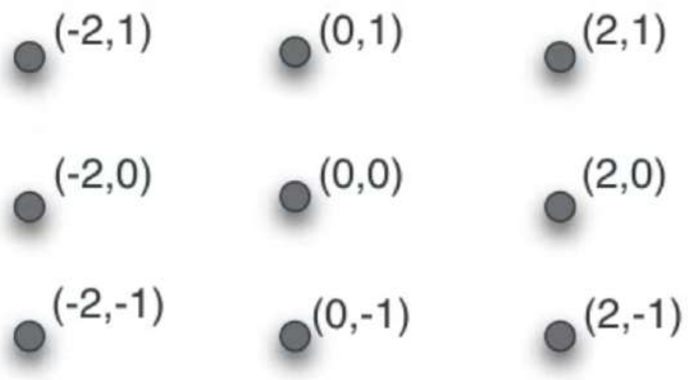
$$\text{covariance} = \frac{(-2) + 0 + (-2)}{3} = -4/3$$

$$\text{covariance} = \frac{2 + 0 + 2}{3} = 4/3$$

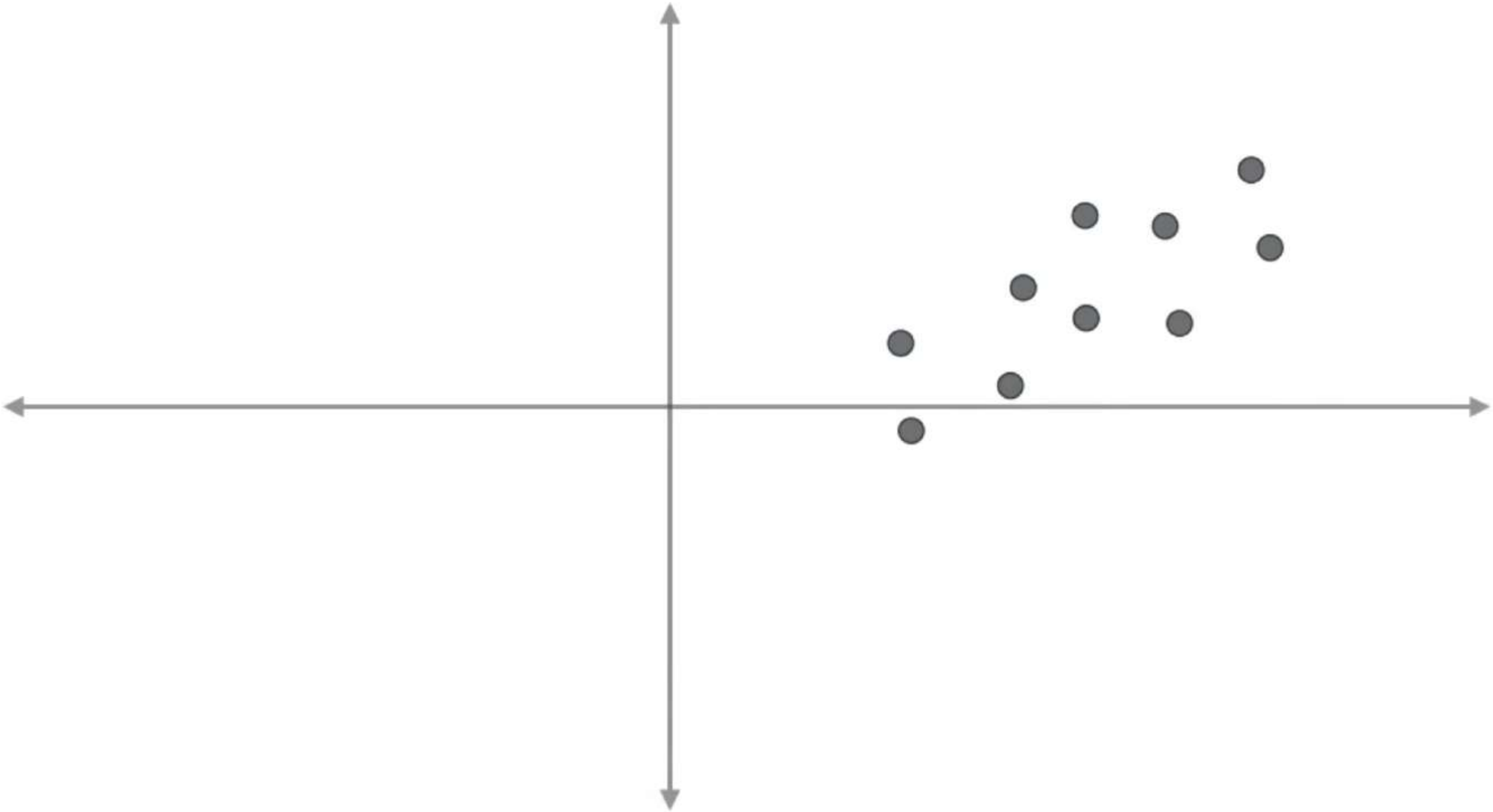
Covariance

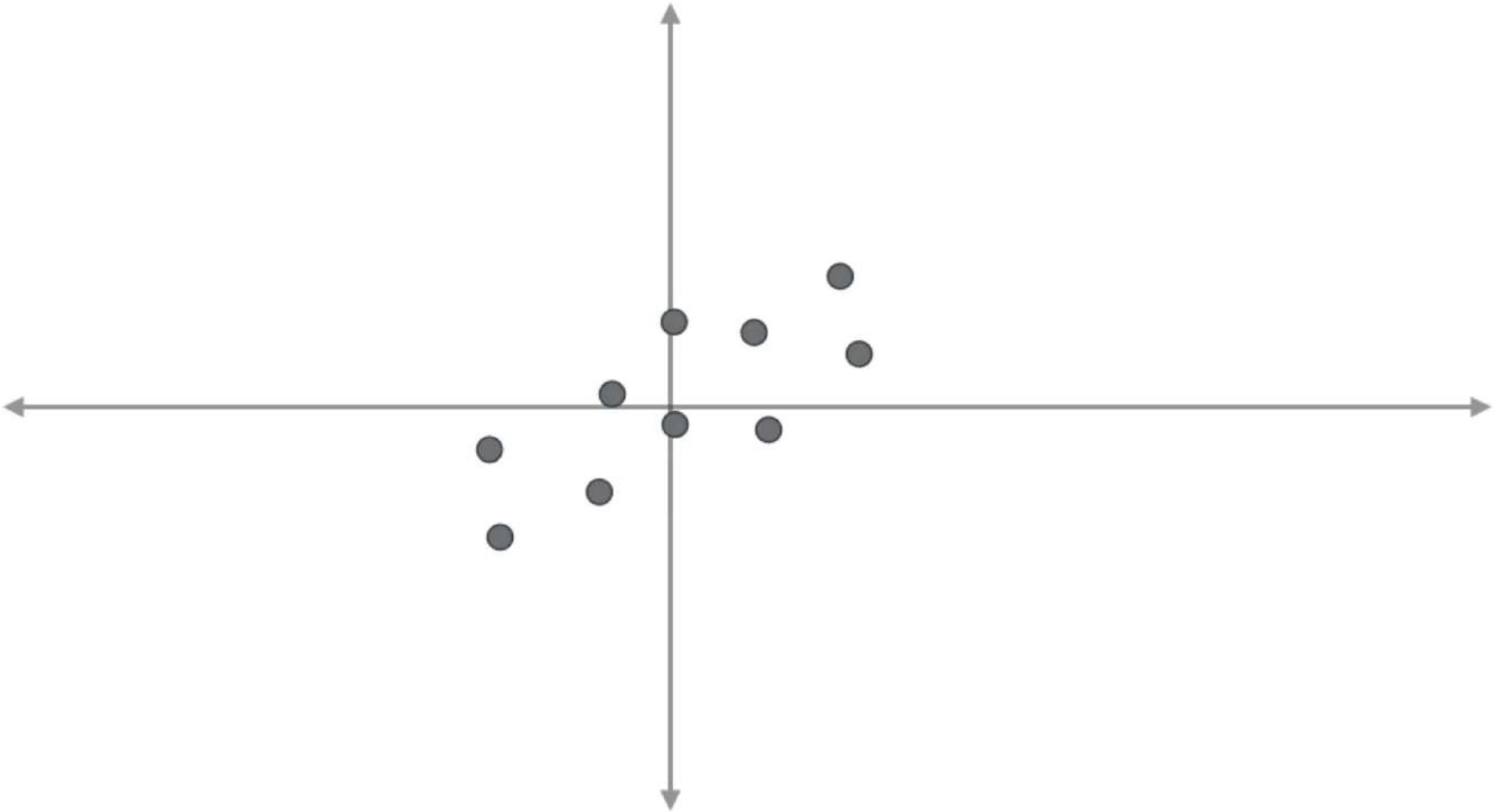


Covariance

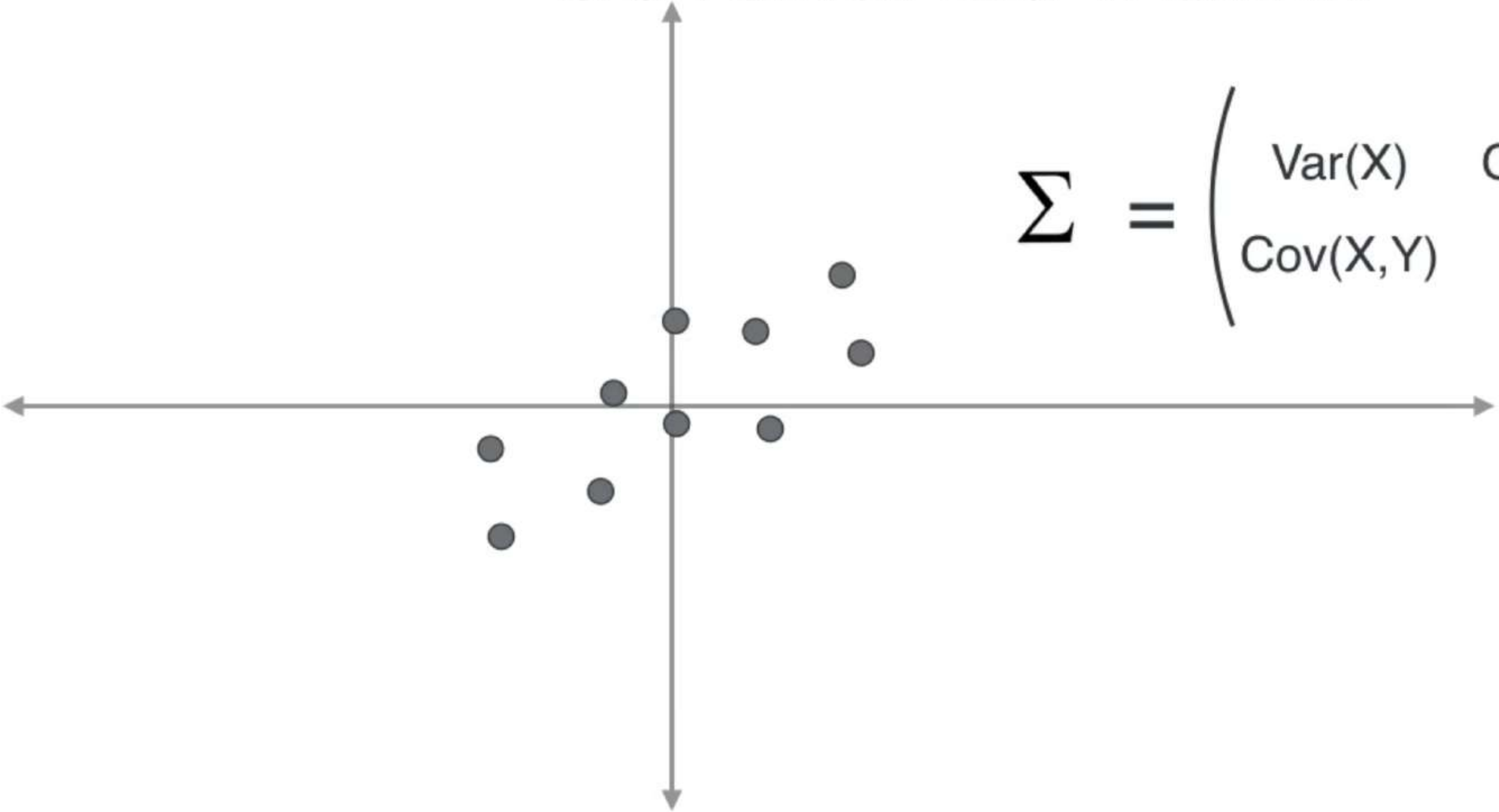


$$\text{covariance} = \frac{-2 + 0 + 2 + 0 + 0 + 0 + 2 + 0 + -2}{9} = 0$$



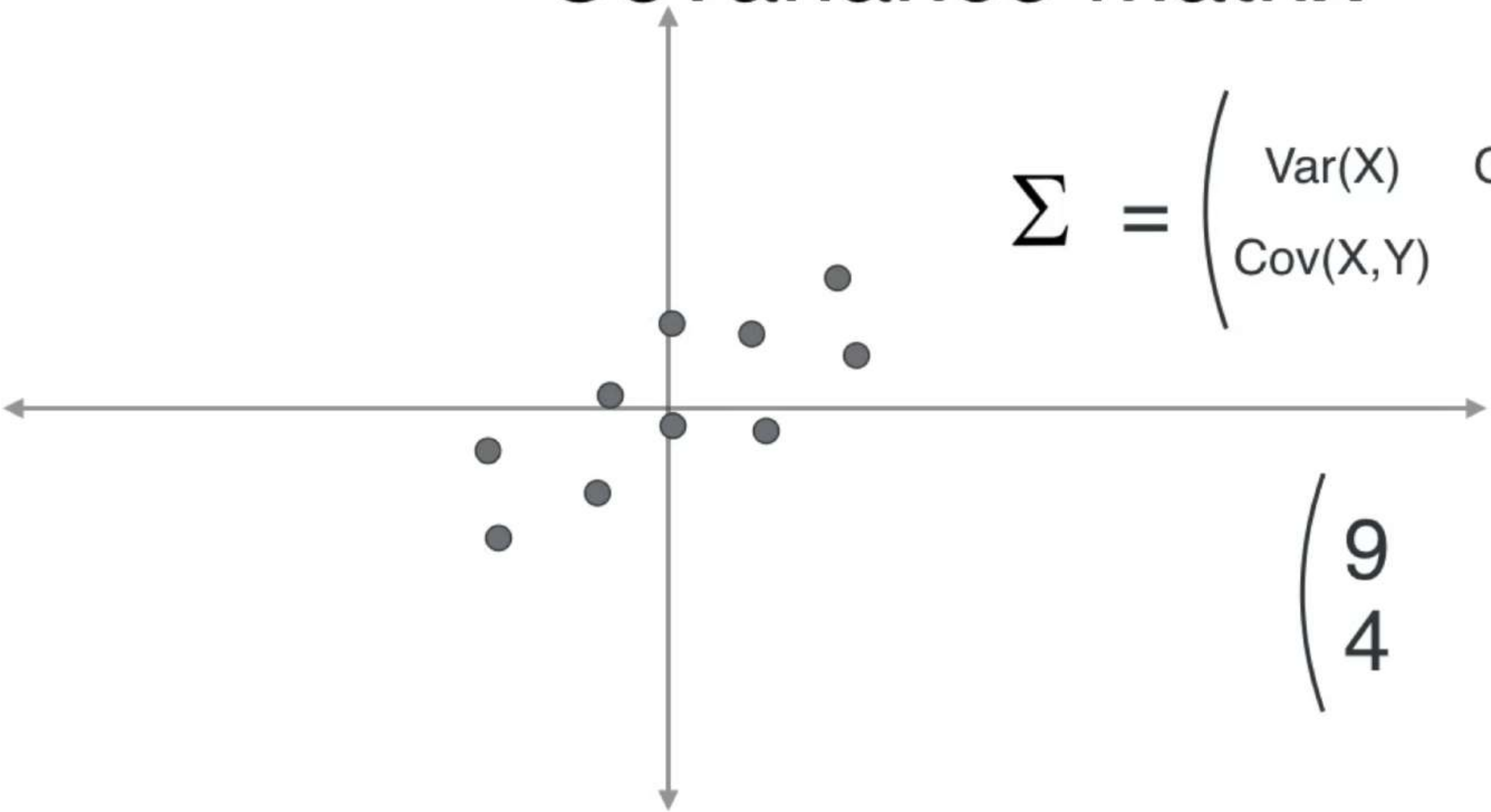


Covariance matrix



$$\Sigma = \begin{pmatrix} \mathrm{Var}(X) & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(X,Y) & \mathrm{Var}(Y) \end{pmatrix}$$

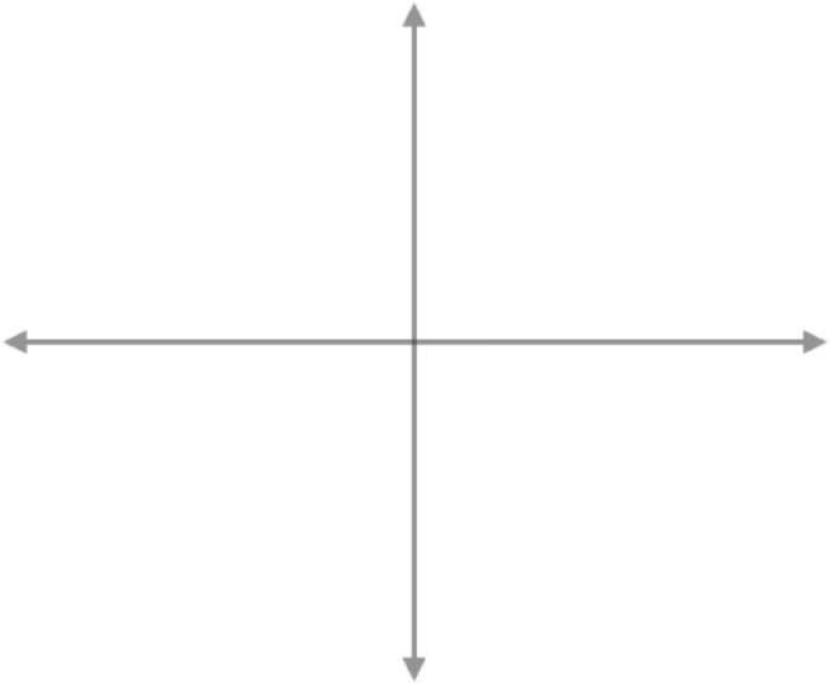
Covariance matrix



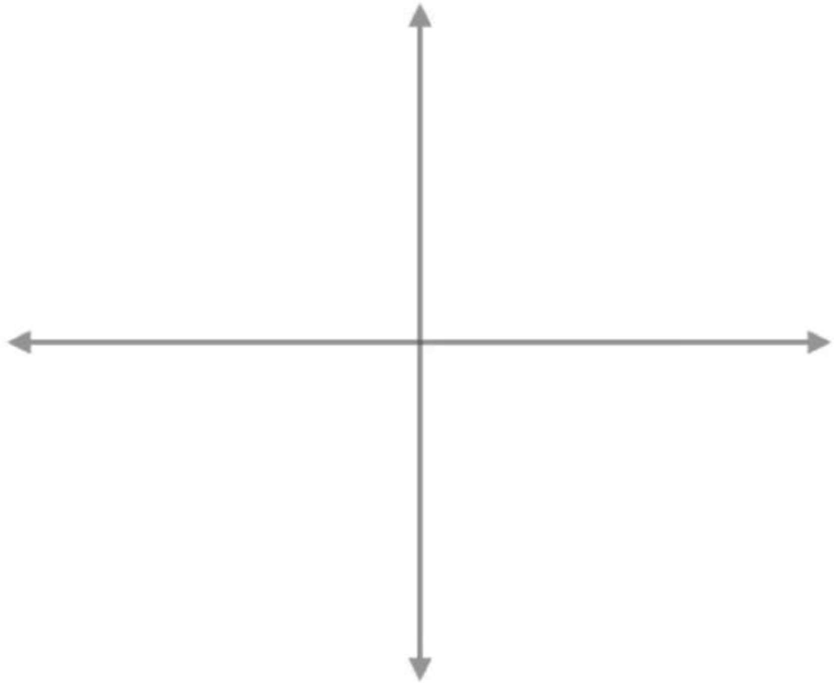
$$\Sigma = \begin{pmatrix} \mathrm{Var}(X) & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(X,Y) & \mathrm{Var}(Y) \end{pmatrix}$$

$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}$$

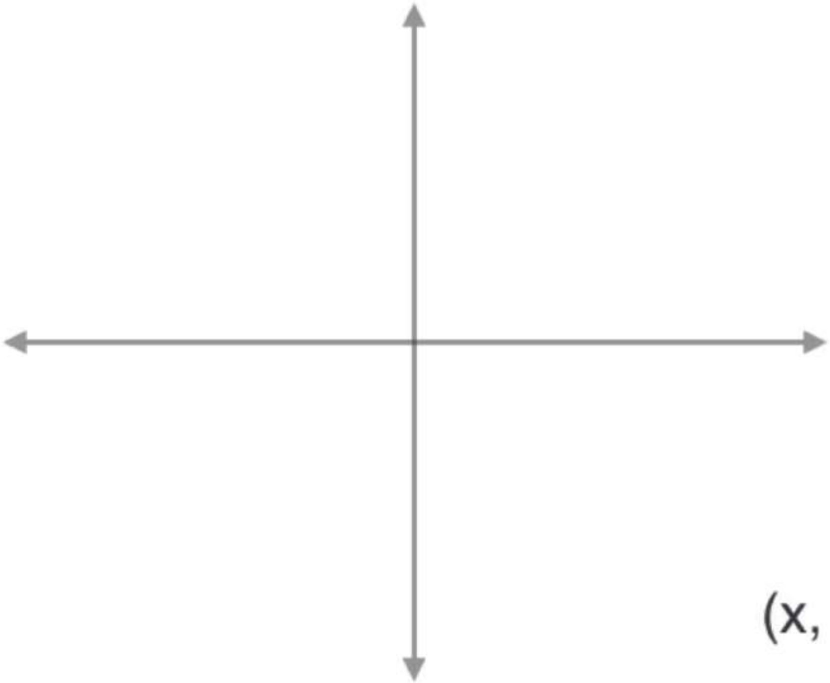
Linear Transformations



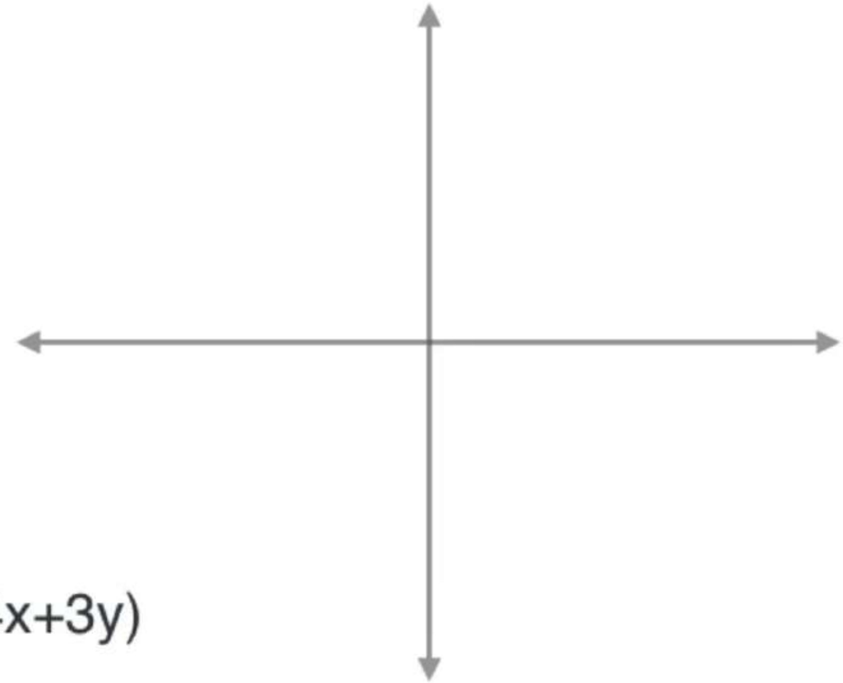
$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix} \longrightarrow$$



Linear Transformations

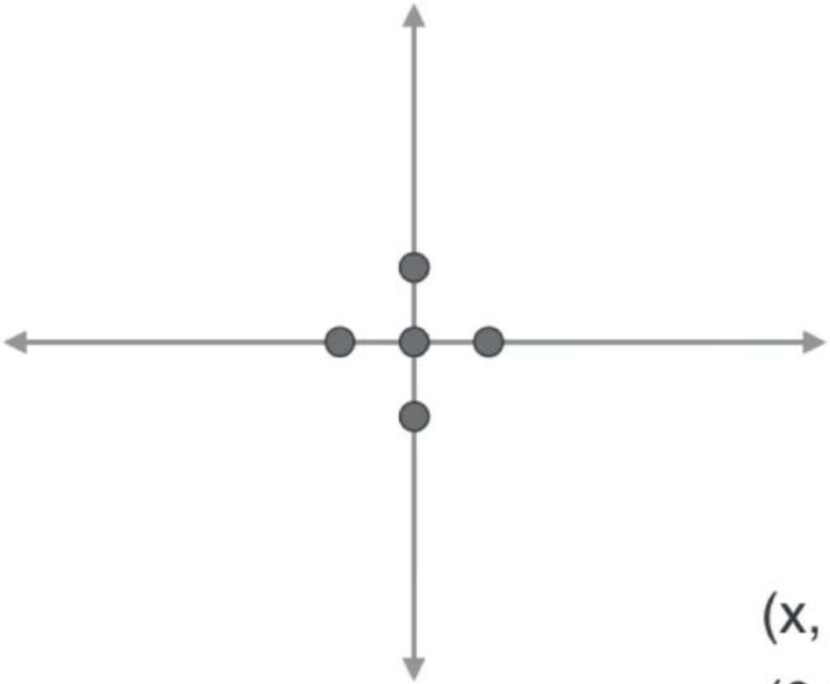


$$\xrightarrow{\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}}$$

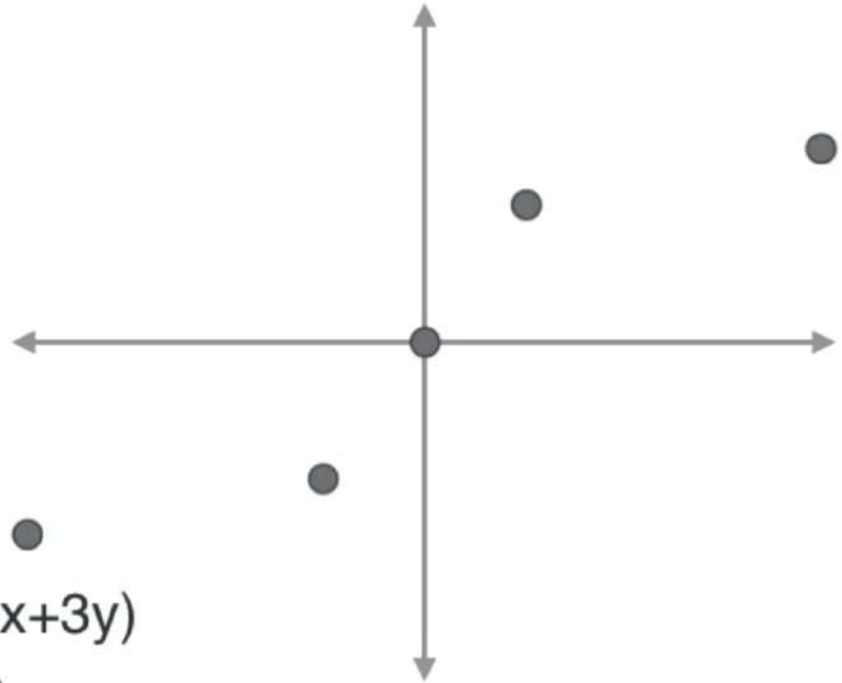


$$(x, y) \longrightarrow (9x+4y,\ 4x+3y)$$

Linear Transformations



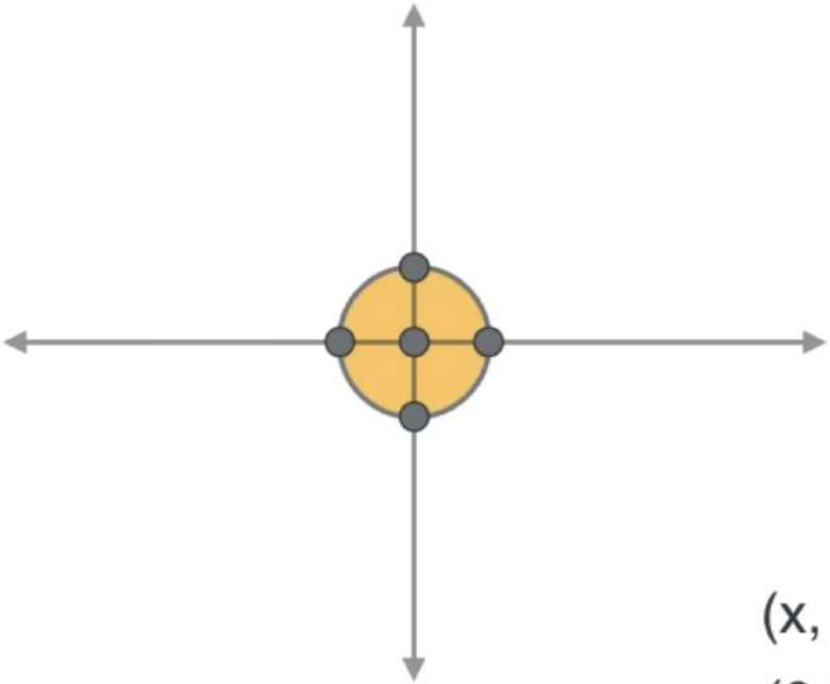
$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix} \longrightarrow$$



$(x, y) \longrightarrow (9x+4y, 4x+3y)$

(0,0)	(0,0)
(1,0)	(9,4)
(0,1)	(4,3)
(-1,0)	(-9,-4)
(0,-1)	(-4,-3)

Linear Transformations



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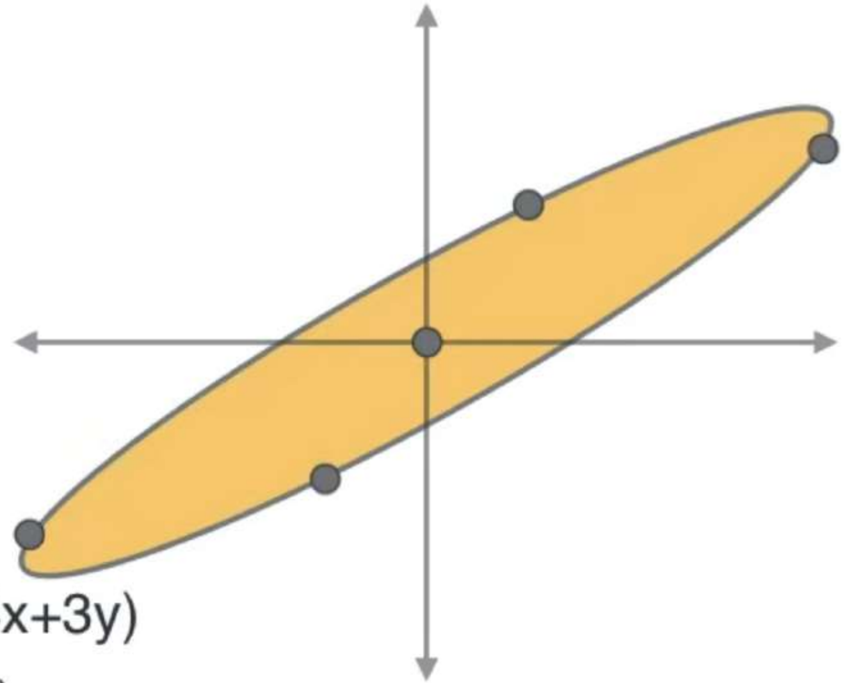

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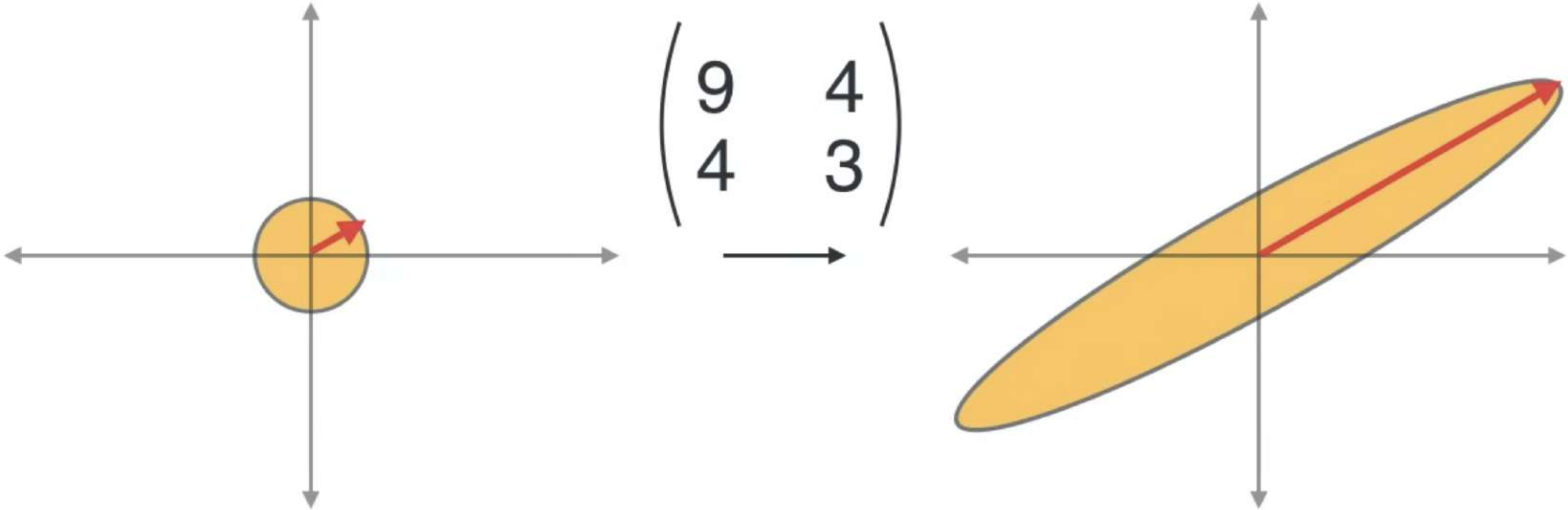
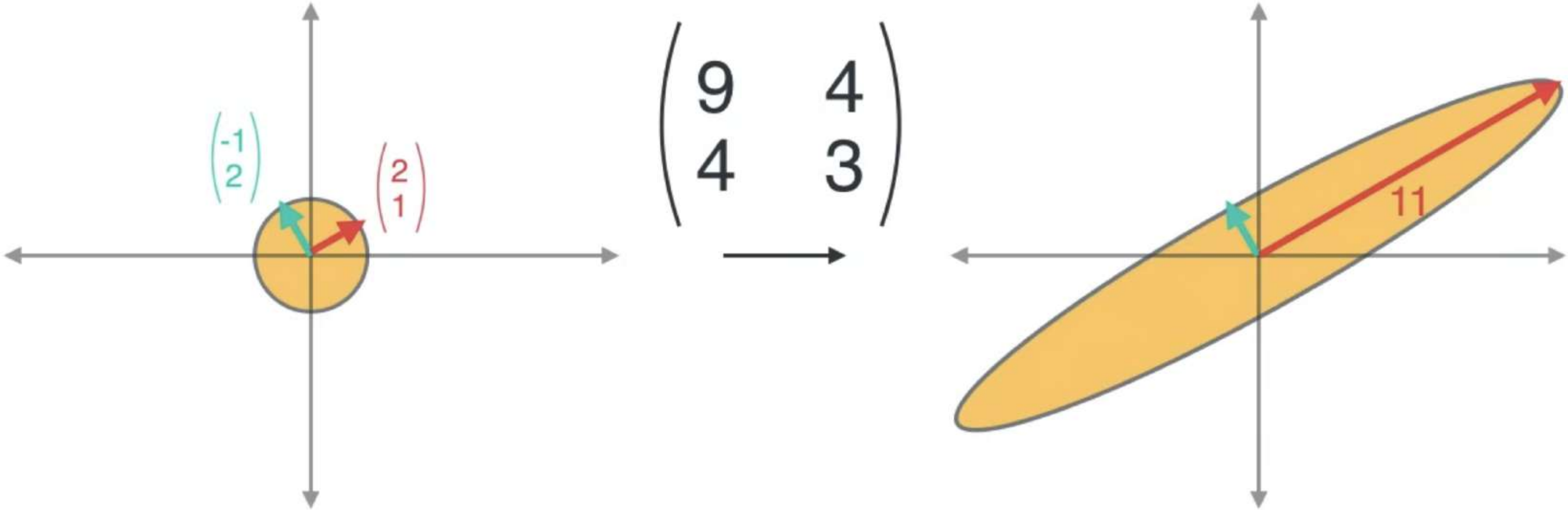
Linear Transformations

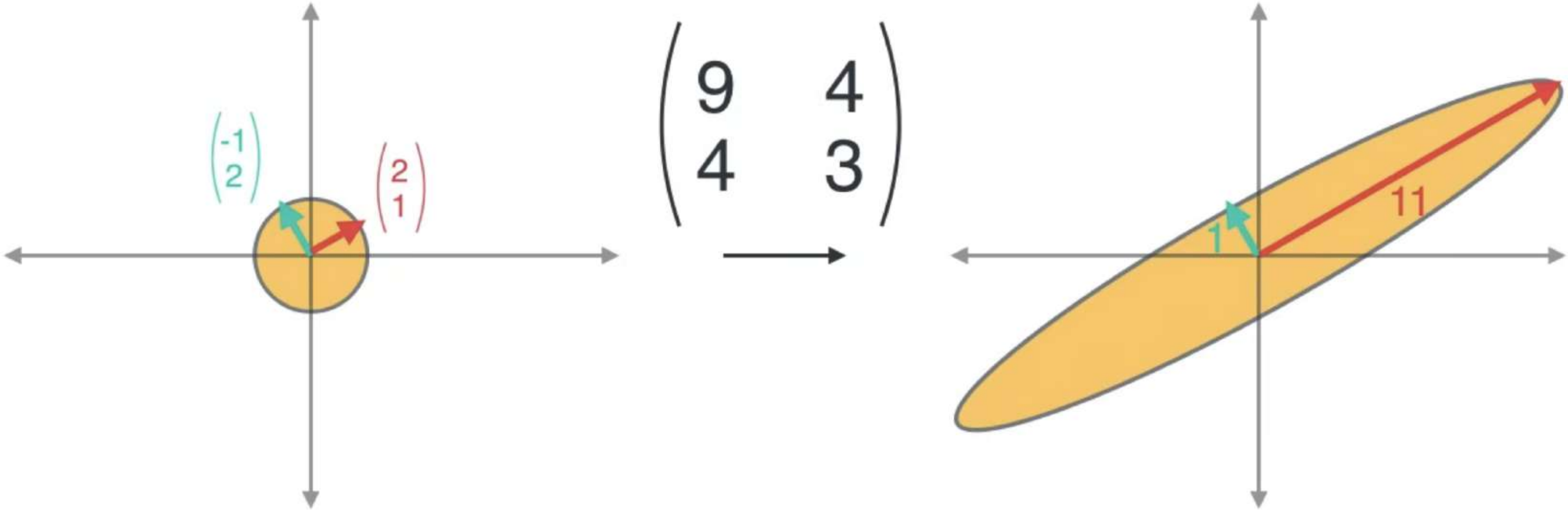
$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix} \longrightarrow$$



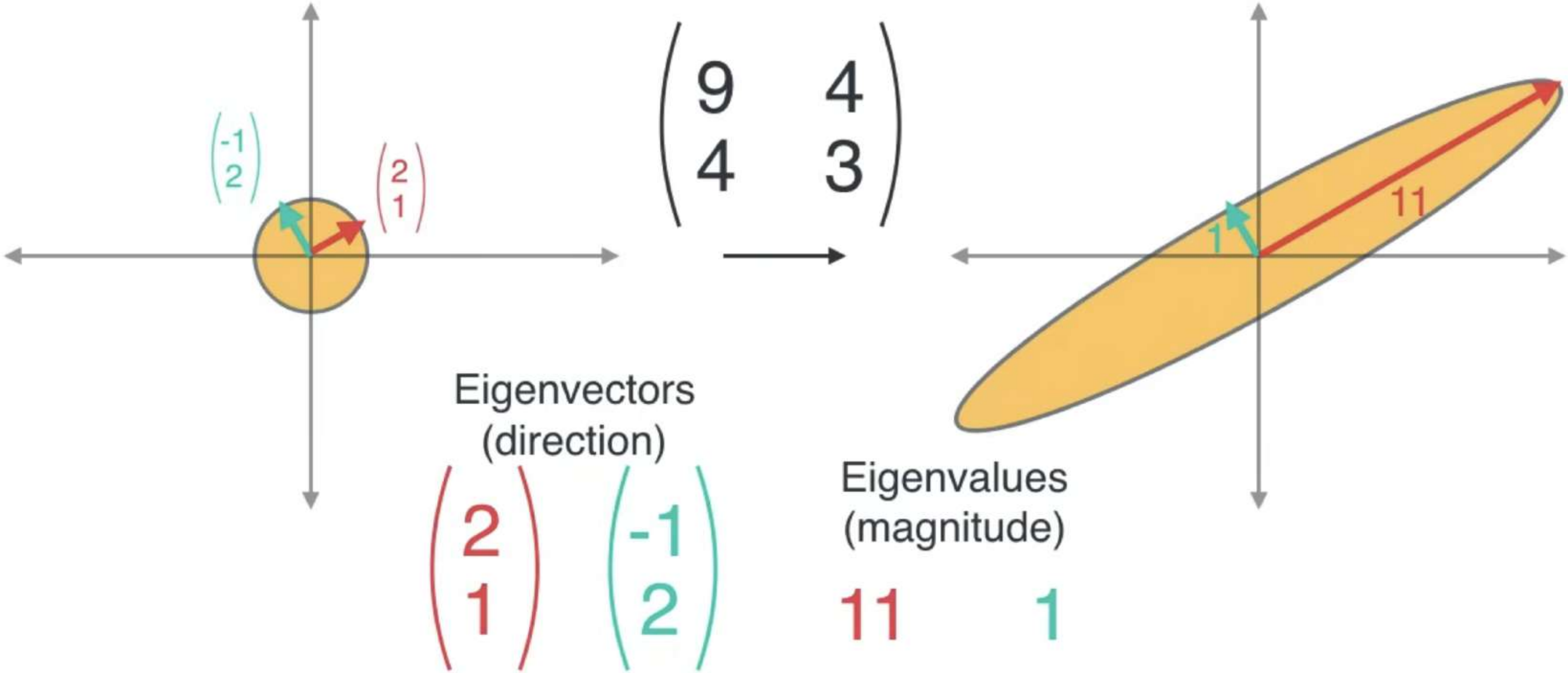
Linear Transformations



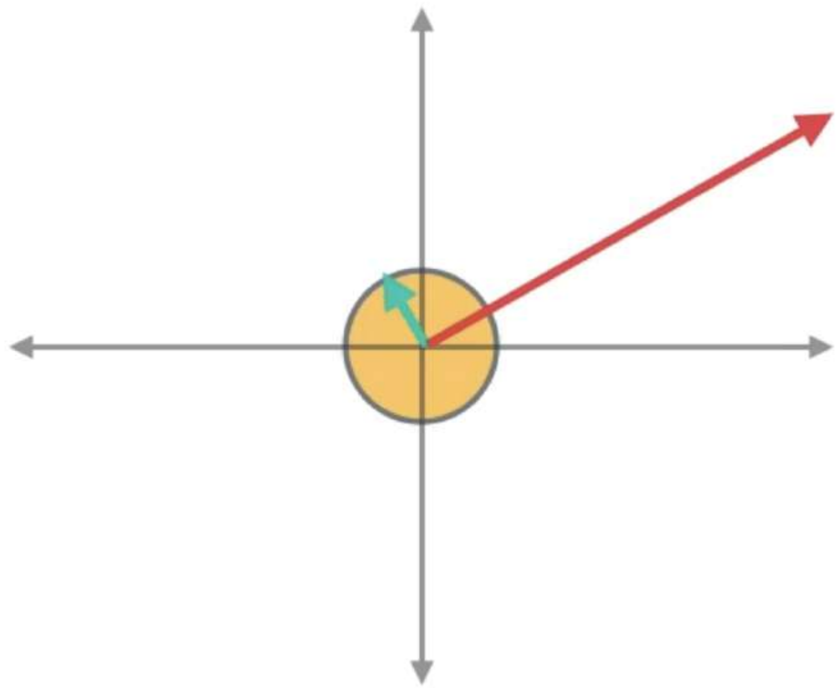
Linear Transformations



Linear Transformations



Linear Transformations



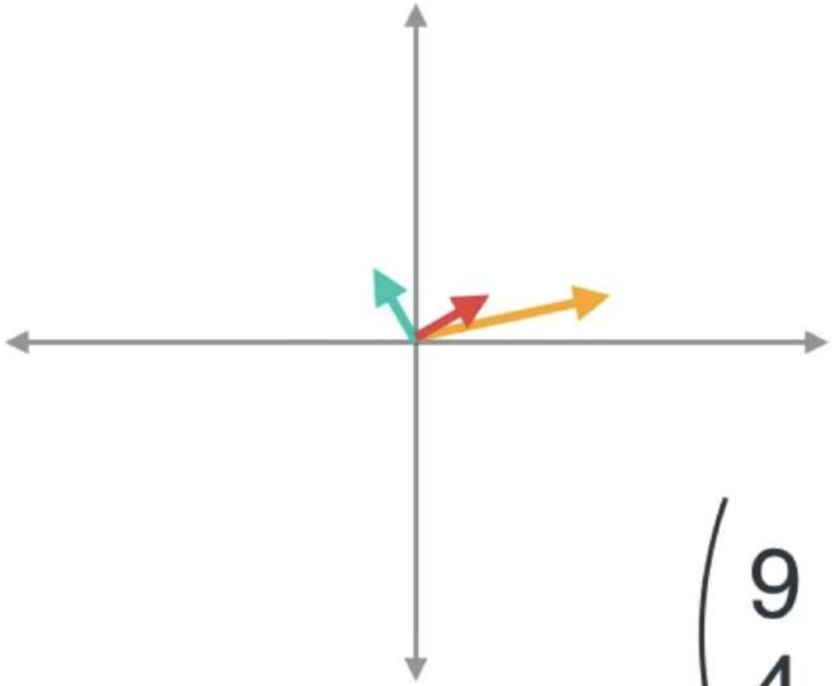
Eigenvectors

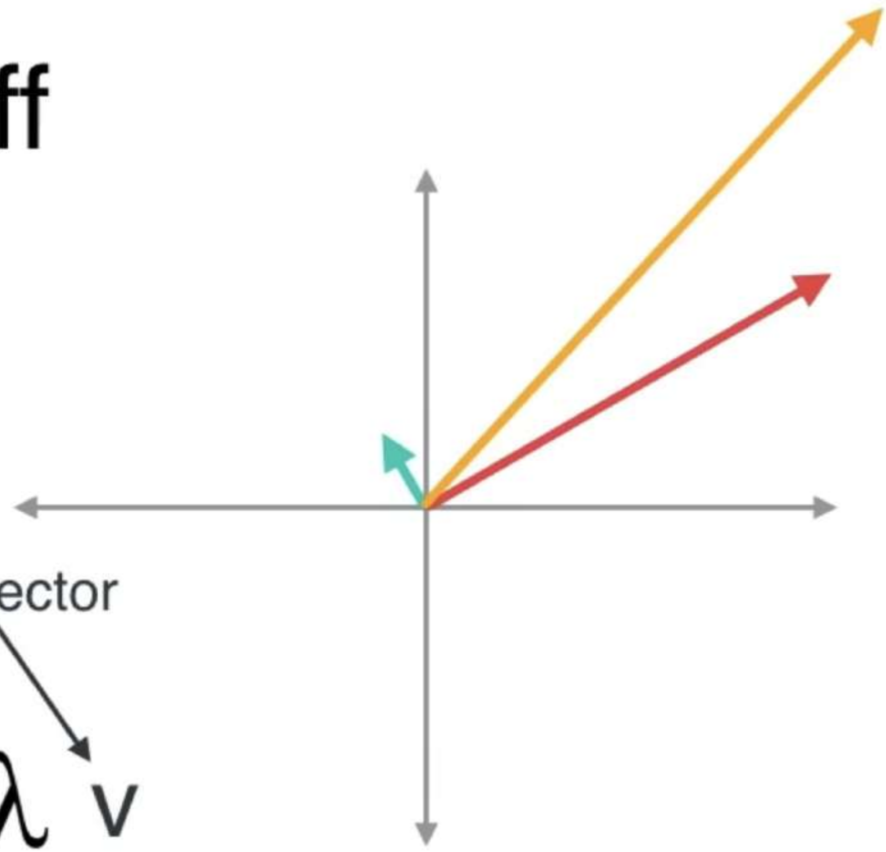
$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$

Eigenvalues

11 1

Eigenstuff





Eigenvector

$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix} v = \lambda \, v$$

Eigenvalue

Eigenvalues

$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}$$

Characteristic Polynomial

$$\begin{vmatrix} x-9 & -4 \\ -4 & x-3 \end{vmatrix} = (x-9)(x-3) - (-4)(-4) = x^2 - 12x + 11$$

$$= (x-11)(x-1)$$

Eigenvalues 11 and 1

Eigenvectors

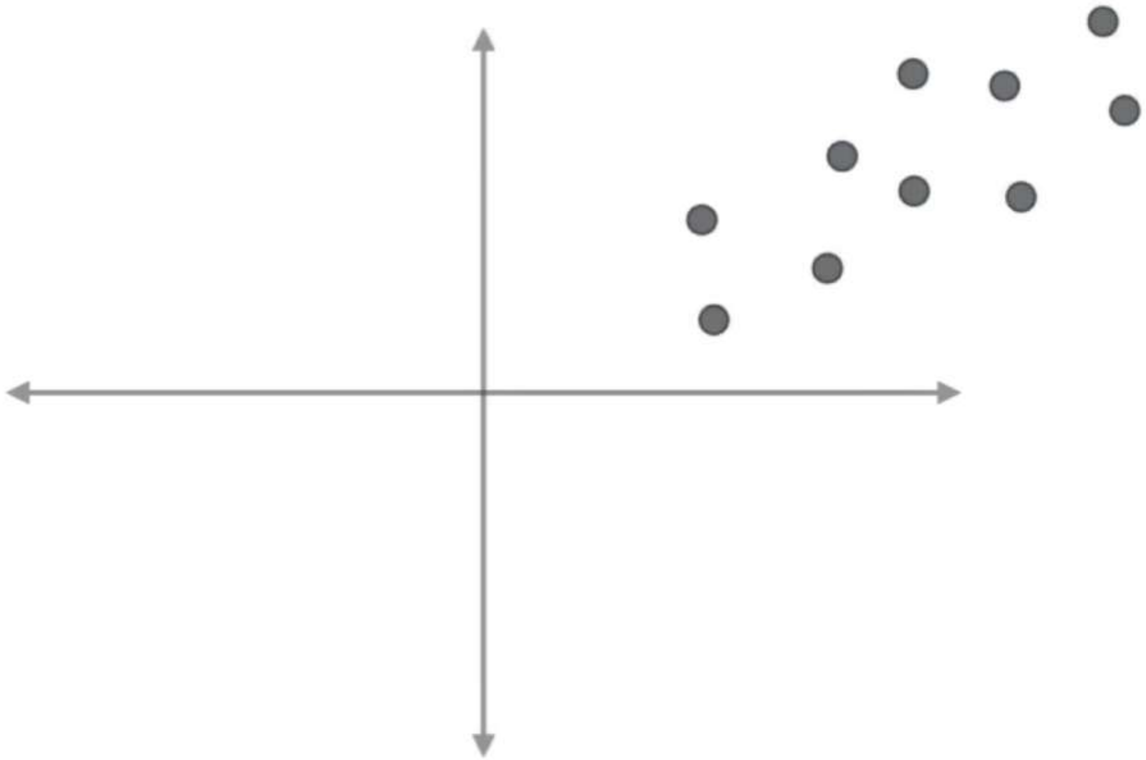
$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = 11\begin{pmatrix} u \\ v \end{pmatrix}$$

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

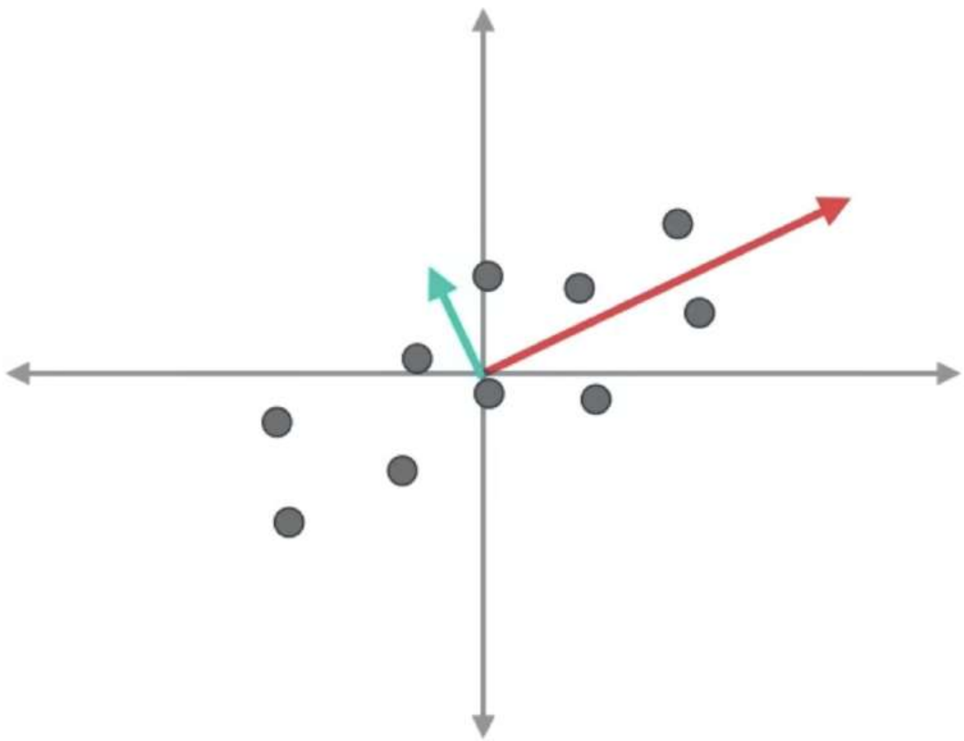
$$\begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = 1\begin{pmatrix} u \\ v \end{pmatrix}$$

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Principal Component Analysis (PCA)



Principal Component Analysis (PCA)



$$\Sigma = \begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}$$

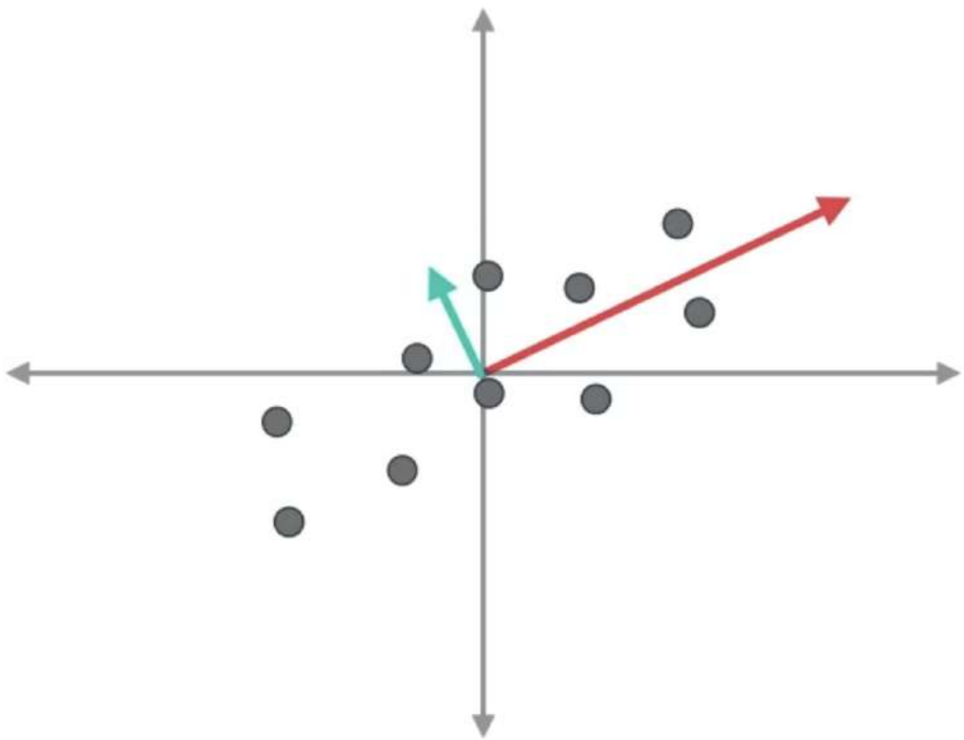
$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Eigenvectors (direction)

$$11 \quad 1$$

Eigenvalues (magnitude)

Principal Component Analysis (PCA)



$$\Sigma = \begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}$$

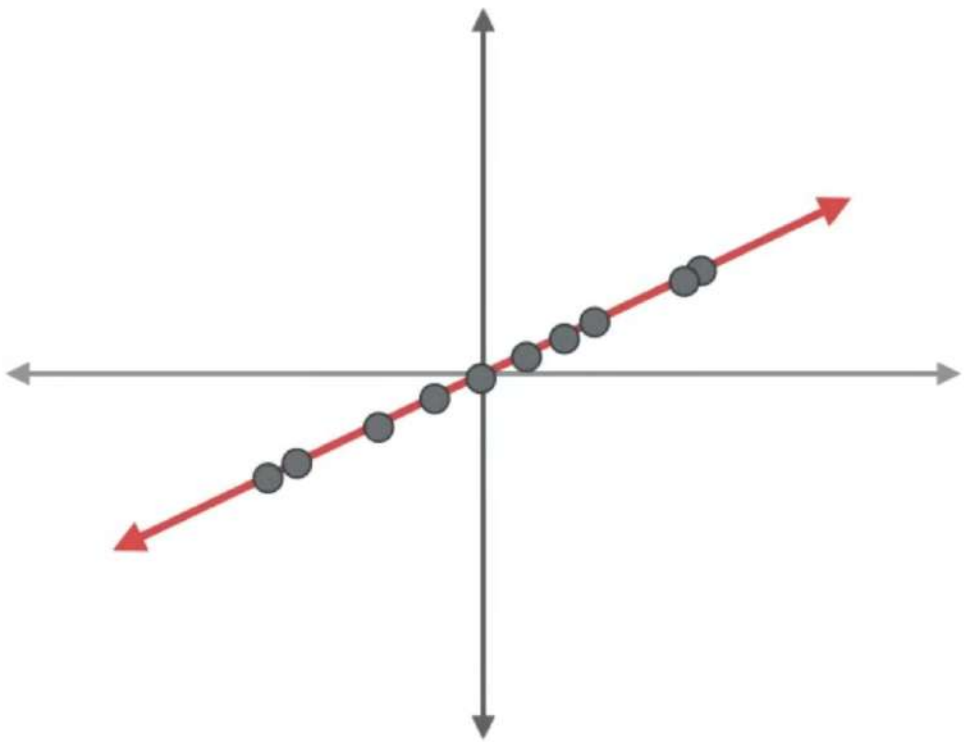
$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Eigenvectors (direction)

$$11 \qquad 1$$

Eigenvalues (magnitude)

Principal Component Analysis (PCA)



$$\Sigma = \begin{pmatrix} 9 & 4 \\ 4 & 3 \end{pmatrix}$$

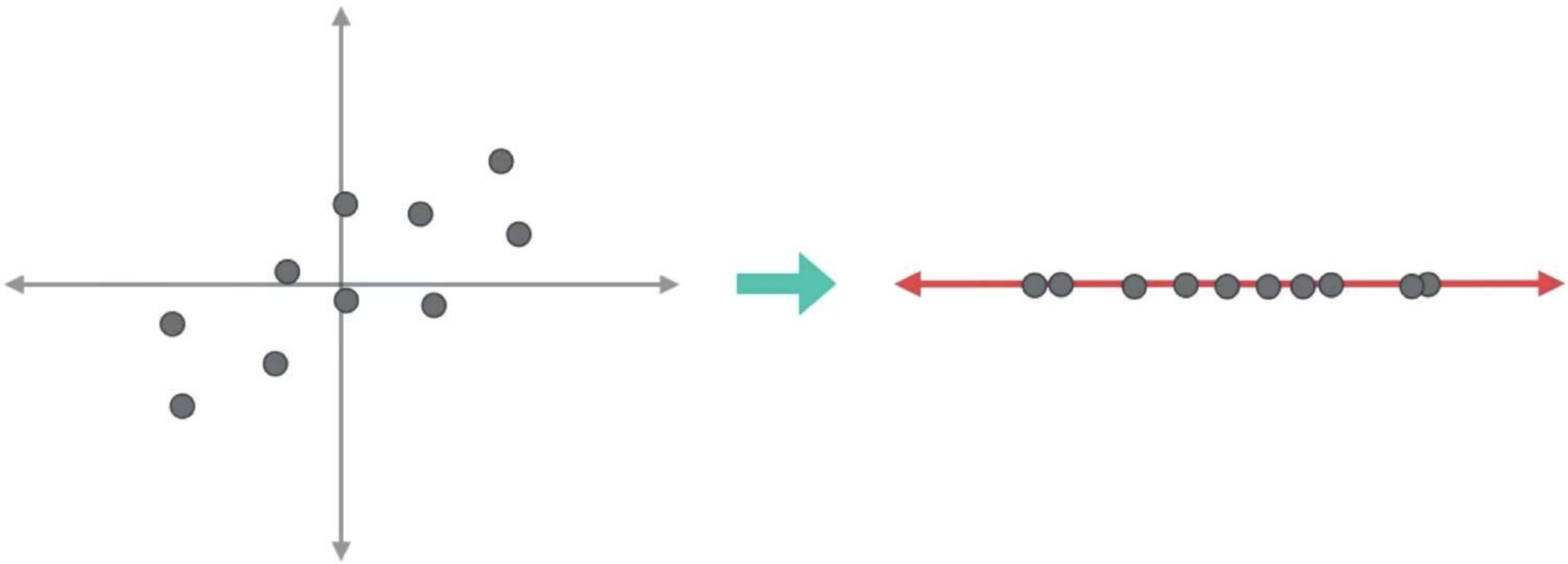
$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Eigenvectors
(direction)

$$11$$

Eigenvalues
(magnitude)

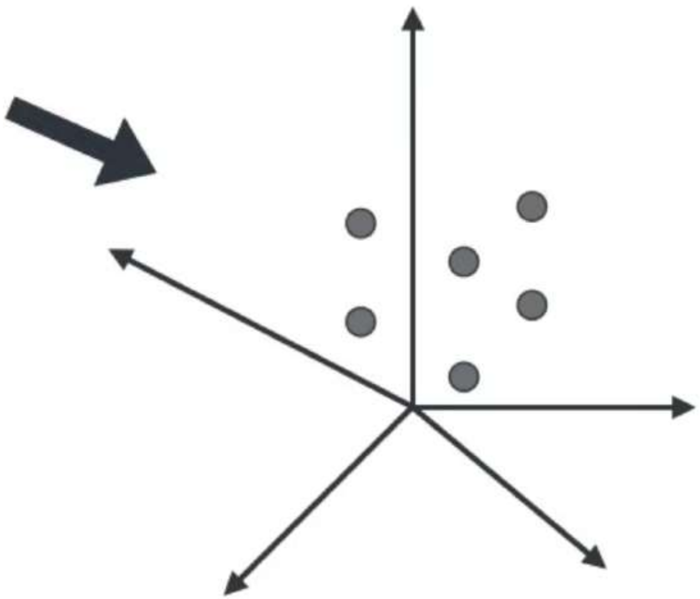
Principal Component Analysis (PCA)



PCA

Large Table

X_1	X_2	X_3	X_4	X_5
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*



5D Plot

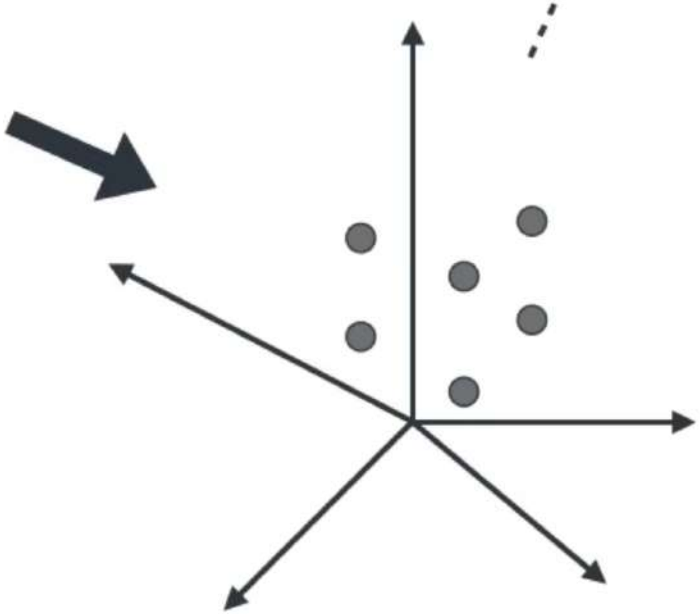
PCA

Covariance matrix

$$\begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix}$$

Large Table

X_1	X_2	X_3	X_4	X_5
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*

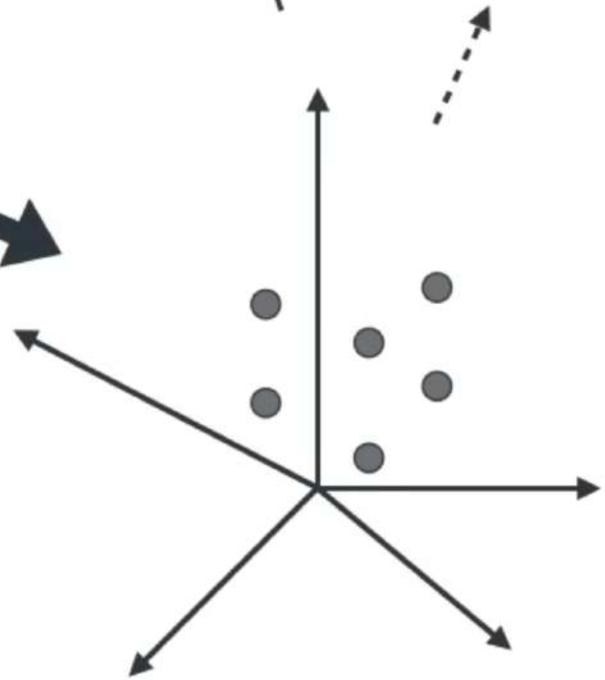


5D Plot

PCA

Large Table

X_1	X_2	X_3	X_4	X_5
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*



5D Plot

Covariance matrix

$$\begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix}$$

Eigenstuff

Big

V_1 λ_1

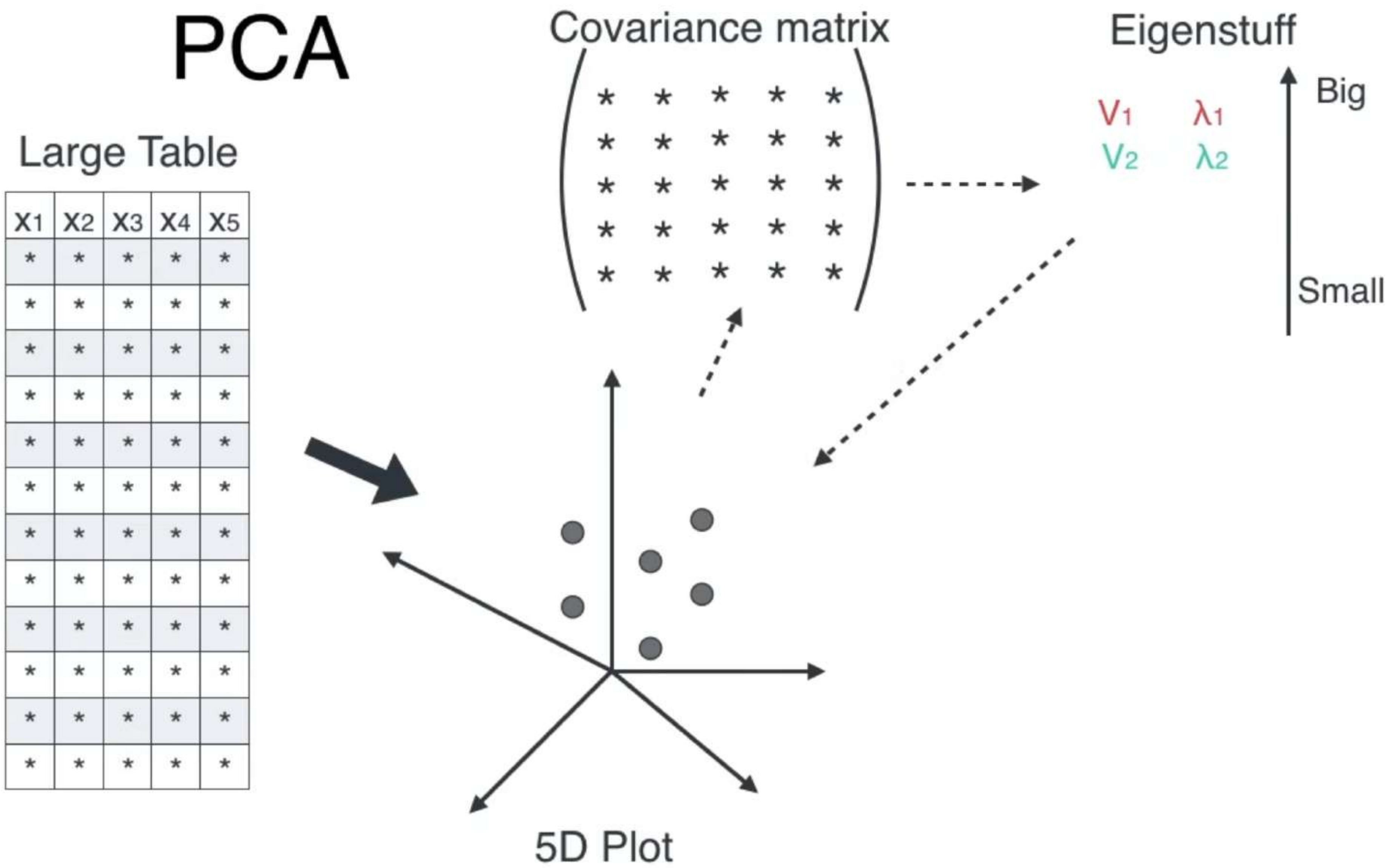
V_2 λ_2

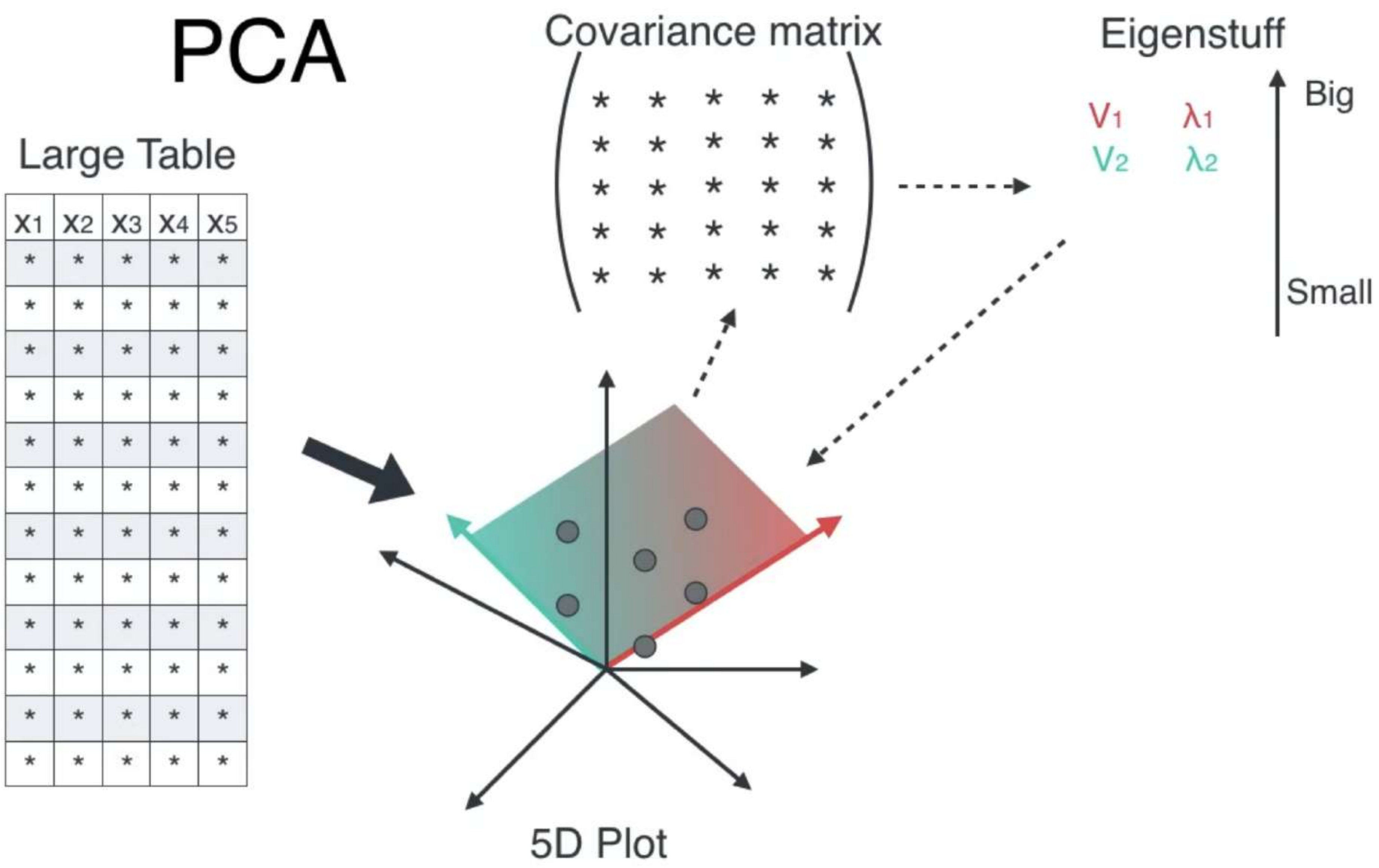
V_3 λ_3

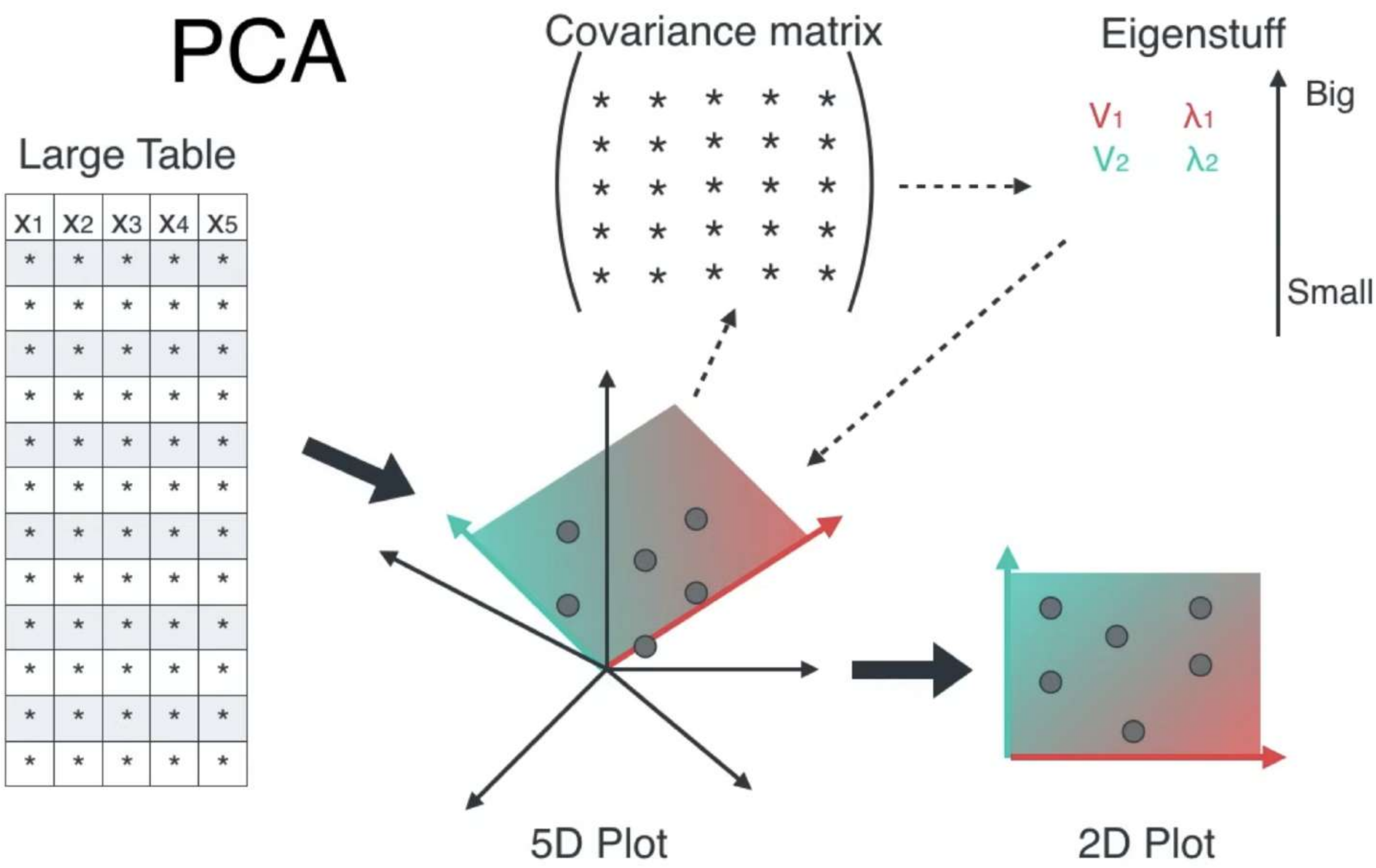
V_4 λ_4

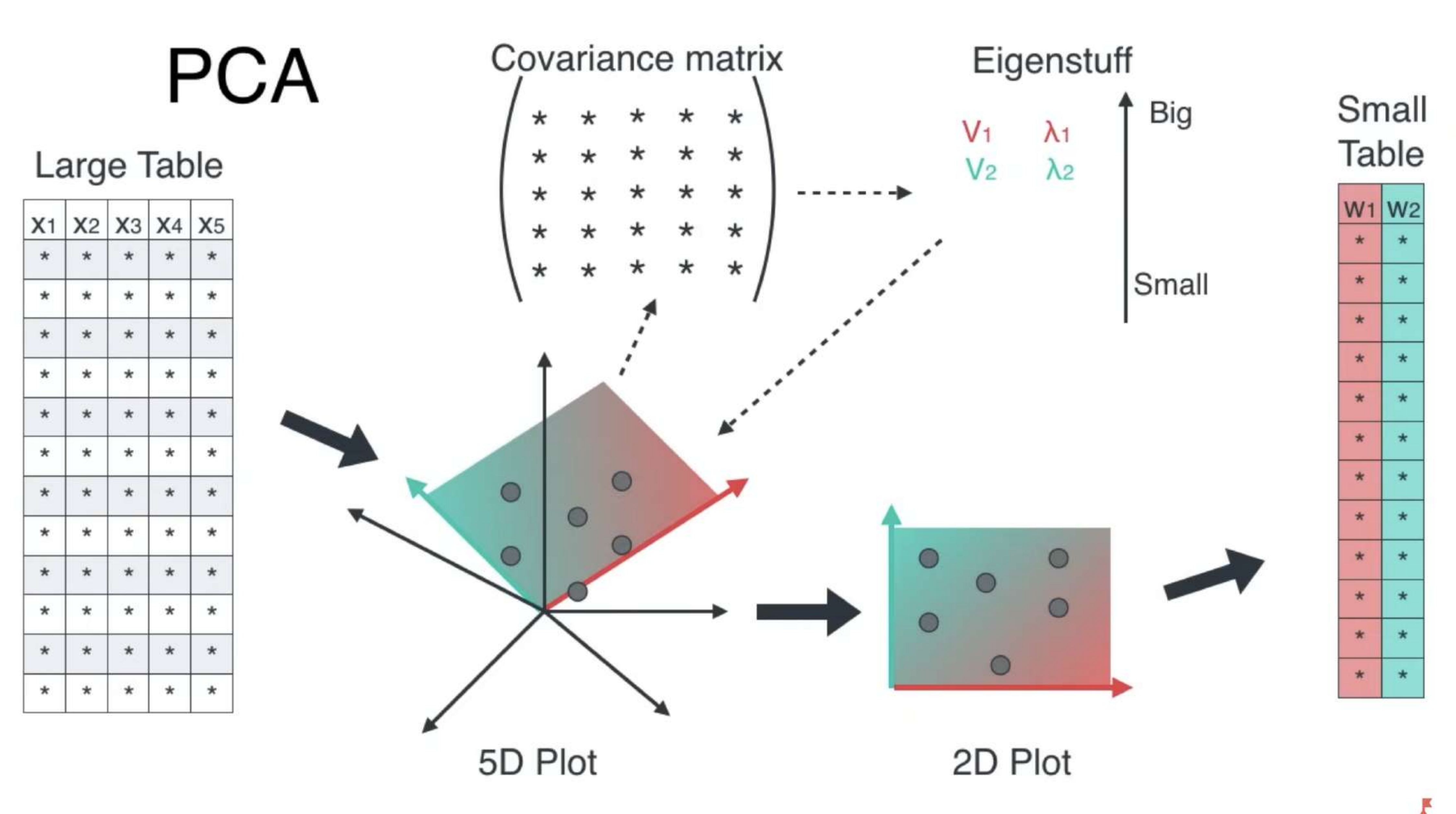
V_5 λ_5

Small


PCA
Covariance matrix
Eigenstuff
Big
V1 λ1
V2 λ2
Small
Large Table
X1 X2 X3 X4 X5
5D Plot


PCA
Large Table
X1 X2 X3 X4 X5
Covariance matrix
Eigenstuff
Big
V1 λ1
V2 λ2
Small
5D Plot


PCA
Large Table
X1 X2 X3 X4 X5
Covariance matrix
Eigenstuff
V1 λ1
V2 λ2
Big
Small
5D Plot
2D Plot


PCA
Large Table
X1 X2 X3 X4 X5
Covariance matrix
Eigenstuff
V1 λ1
V2 λ2
Big
Small
Small Table
W1 W2
5D Plot
2D Plot